

Distributional Weak σ —Convergence and Quantile Treatment Effects

Abhishek Pandey* and Donggyu Sul†

May, 2026: Incomplete DO NOT CITE

Abstract

This paper proposes a pre-estimation diagnostic for quantile treatment effects on the treated (QTT). Quantile difference-in-differences (QDID) and QTT methods assume that the control group’s distribution credibly represents the counterfactual distribution over time. In practice, this assumption is usually untested. We employ weak and distributional σ -convergence tests to determine whether the control distribution remains non-divergent around its mean and relevant quantiles. If the control distribution fails these diagnostics, the proposed procedure applies a sieving algorithm to construct a distributionally non divergent control group before QTT estimation. Once a suitable control group is established, the procedure tests whether the treated distribution changes only by a common location shift or if treated-control quantile gaps diverge over time. This diagnostic informs researchers when an average treatment effect on the treated (ATT) interpretation is defensible and when direct QTT estimation is necessary to understand the sign, magnitude, and location of effects across the distribution.

Keywords: distributional treatment effects, quantile treatment effects, difference-in-differences, parallel trends, σ -convergence.

JEL Classification: C21, C23, C31, C33.

*Department of Economics, University of Texas at Dallas, Email: axp220266@utdallas.edu

†Department of Economics, University of Texas at Dallas, Email: d.sul@utdallas.edu.

1 Introduction

Quantile treatment effects on the treated (QTT) have received increasing attention in applied work, particularly when the average treatment effects on the treated (ATT) are not statistically significant or very small. The ATT may be small or statistically insignificant even when a policy changes the lower tail, the upper tail, or the dispersion of the treated outcome distribution. In such cases, the mean difference between treated and control groups gives an incomplete description of the treatment effect. QTT is attractive because it compares entire outcome distribution between the treated and control groups, it provides more detailed and comprehensive information on how a treatment influences the treated group.

The econometric literature provides powerful tools for distributional DID and QTT under changes-in-changes restrictions, distributional parallel trends, copula stability, and panel quantile restrictions (Athey and Imbens, 2006; Callaway and Li, 2019; Callaway, Li, and Oka, 2018; Chernozhukov, Fernández-Val, and Melly, 2013; Kim and Wooldridge, 2025). Yet QTT remains less developed than ATT in staggered DID settings. For mean effects, recent work provides estimators and diagnostics for staggered adoption, heterogeneous effects, covariate adjustment, and sensitivity to parallel-trends violations (Borusyak, Jaravel, and Spiess, 2024; Callaway and Sant’Anna, 2021; Roth et al. , 2023; Sant’Anna and Zhao, 2020). For quantile effects, Li and Lin (2024) and Galvao and Miller (2025) are important recent advances, showing how QTT can be identified and estimated with staggered treatment timing answering the estimation question under credible distributional identifying assumptions. This paper asks the prior diagnostic question: is the control group distribution diverging over time, and is QTT empirically valid?

Estimating QTT requires the identifying distributional assumption to hold. These assumptions won’t be valid if the distribution of control group diverges over time. This issue is especially important in non-experimental social-science panels. In a randomized medical trial, the control and treated groups are often designed to have similar baseline distributions before treatment. In social-science applications, by contrast, control units may have different demographics, different local shocks, and different exposure to aggregate trends. Even when average outcomes satisfy a parallel-trends restriction, the cross-sectional variance or the shape of the control distribution may diverge over time. If the control distribution itself is diverging, treated-control quantile comparisons may reflect instability in the control group distribution rather than treatment-induced distributional effects .

The purpose of this paper is threefold. First, we propose a new definition of distributional σ -convergence and a corresponding test to determine whether the untreated units exhibit

distributional divergence. Weak σ -convergence, developed by [Kong, Phillips, and Sul \(2019\)](#), asks whether the cross-sectional dispersion of an outcome around its mean remains bounded over time. We extend this idea to quantiles. An outcome is distributionally non-divergent if the dispersion of the outcome around its quantiles, cross-sectional mean of the squared difference between the outcome and its quantiles, does not diverge over time. Statistically, the concept of distributional stability or non-divergence is more flexible than that of identical distribution. We also propose a formal statistical test to examine the null hypothesis of no distributional divergence.

Second, when the control distribution fails the non-divergence test, we use a sieving procedure to construct a non-diverging control group before QTT is estimated. The logic follows the sieving methods of [Phillips and Sul \(2007\)](#), [Lee et al. \(2026\)](#) and [Lee and Sul \(2026\)](#) adapted to the distributional setting. Unlike previous sieving methods, which focuses on the movement of outcomes relative to a central tendency, our method considers the overall dynamics between outcomes and various quantiles. Consequently, this new sieving method ensures that the distributions of both controlled and ever-treated units are similar, a much stronger restriction than the parallel trends assumption.

Third and the most important, by utilizing the notion of distributional sigma convergence, we design a formal pre-test to determine whether applied researchers should consider the QTT or simply report the ATT estimation, conditional on a non-diverging control group. If the treated distribution remains non-diverging and the ATT is constant, the treatment effect has a common-shift interpretation, so the ATT is an adequate empirical summary. If the treated group has σ divergence and treated distributional divergence or treated-control quantile distances diverge, then the average effect is not a complete description of the policy effect and QTT should be estimated at the affected quantiles. This last step is a diagnostic, not a signed QTT estimator. The statistic is based on squared distances. Rejection therefore indicates quantile separation, but the sign and magnitude of QTT must be obtained from direct QTT estimation.

The contribution of the paper is therefore not to introduce another estimator of QTT. It is to provide a pre-estimation diagnostic and classification procedure for distributional DID. Existing QTT-QDID methods begin after the researcher has accepted the distributional identifying assumptions. This paper places an observable diagnostic before that step. It tests whether the control group can serve as a credible counterfactual, constructs a non-diverging control group when necessary, and then determines whether the data call for direct QTT estimation or whether ATT is sufficient for the empirical question.

The rest of the paper is organized as follows. Section 2 sets up the notation and defines

weak and distributional σ -convergence. Section 3 gives the asymptotic justification for the slope tests. Section 4 presents the classification. Section 5 describes the testing procedure. Section 6 reports Monte Carlo evidence. Section 7 presents the empirical illustration. Section 8 discusses interpretation and reporting. Section 9 concludes. Proofs are collected in the appendix.

2 Preliminary

This section introduces the notation, definitions, and key assumptions. We then discuss the relationship between the distributional convergence test and quantile treatment effects.

2.1 Notations and Definitions

Let $i = 1, \dots, n$ index the units and $t = 1, \dots, T$ index the time periods. Let $D_{it} \in \{0, 1\}$ denote treatment status, with two groups defined: the never-treated or control group (indexed by N) and the ever-treated group (indexed by I). Treatment for the ever-treated group begins at time $T_0 + 1$ and continues until time T . Note that the ever-treated units were not treated until $t = T_0$. This basic framework can be extended to accommodate staggered treatment timings across different units. The number of units in the control group and the treated group are denoted as n_N and n_I , respectively, with the relationship $n_N + n_I = n$.

Let $\mu_{N,t}$ denote the mean of non-treated units in the control group, and let $\mu_{N,t}^\alpha$ denote the corresponding α -quantile with $0 < \alpha < 1$. Define $\mu_{I,t}$ and $\mu_{I,t}^\alpha$ analogously for the treated group. Next, denote the control mean-to-quantile distance as $d_{N,t}^\alpha = |\mu_{N,t} - \mu_{N,t}^\alpha|$. Similarly, denote $d_{I,t}^\alpha = |\mu_{I,t} - \mu_{I,t}^\alpha|$.

Next, we introduce the concept of weak σ -convergence developed by [Kong, Phillips, and Sul \(2019\)](#).

Definition 1 (Weak σ -Convergence) *A panel series y_{it} is weakly σ -convergent towards its cross-sectional mean μ_t if the following conditions hold:*

- (a) $\text{plim}_{n \rightarrow \infty} n^{-1} \sum_{i=1}^n (y_{it} - \mu_t)^2 = K_t < \infty$ a.s. for all t ;
- (b) $\text{plim}_{t \rightarrow \infty} K_t \in [0, \infty)$;
- (c) $\limsup_{T \rightarrow \infty} c_T^{-1} \sum_{t=1}^T \tilde{K}_t \tilde{t} < 0$ a.s. for some increasing sequence $c_T \rightarrow \infty$ as $T \rightarrow \infty$, where $\tilde{z}_t = z_t - T^{-1} \sum_{t=1}^T z_t$ denotes the demeaned series of z_t .

Throughout the paper, we need the notion of no σ -divergence rather than weak σ -convergence. When $\limsup_{T \rightarrow \infty} c_T^{-1} \sum_{t=1}^T \tilde{K}_t \tilde{t} \leq 0$, we say that y_{it} is weakly σ non-divergent.

Both the weak σ -convergence and non-divergent are performed by running the following simple linear trend regression.

$$K_t = \phi_0 + \phi t + u_t, \quad (1)$$

where K_t is the estimated cross-sectional variance of y_{it} , that is, $K_t = n^{-1} \sum_{i=1}^n (y_{it} - n^{-1} \sum_{i=1}^n y_{it})^2$. The t -ratio of $\hat{\phi}$ is constructed as

$$t_{\hat{\phi}} = \frac{\hat{\phi}}{\sqrt{\hat{\Omega}_u^2 / \sum_{t=1}^T \tilde{t}^2}}, \quad (2)$$

with $\tilde{t} = t - T^{-1} \sum_{t=1}^T t$. Note that

$$\hat{\Omega}_u^2 = \frac{1}{T} \sum_{t=1}^T \hat{u}_t^2 + \frac{2}{T} \sum_{t=1}^T \sum_{j=1}^L \left(1 - \frac{j}{L+1}\right) \hat{u}_t \hat{u}_{t-j} \quad (3)$$

with $L = \lfloor T^\kappa \rfloor$ for $\kappa = 1/3$. If y_{it} converges to its cross-sectional mean, then $t_{\hat{\phi}} \rightarrow -\infty$ as $n, T \rightarrow \infty$ and $T/n \rightarrow 0$. This indicates weak σ -convergence, according to [Kong, Phillips, and Sul \(2019\)](#). When y_{it} fluctuates around its cross-sectional mean, the t -ratio can be approximated by a standard normal distribution. Conversely, if y_{it} diverges from its mean, the t -ratio diverges to positive infinity. Hence, weak σ non-divergence can be tested at the threshold point of 1.65.

[Lee and Sul \(2026\)](#) apply this concept to test the parallel trends assumption. If y_{it} does not share a common trend, cross-sectional dispersion must increase over time. They demonstrate that even if only one unit exhibits a different trend, the t -ratio will be greater than 1.65. As technological progress accelerates and physical distance becomes less significant, cross-dependence across regions increases. This convergence is one reason why panel data—including income, wages, life expectancy, and crime rates—shows a trend toward convergence over time.

There are two reasons why panel data may diverge from its cross-sectional average. Consider the following data-generating process for an outcome variable, y_{it} :

$$y_{it} = a_i + b_i \theta_t + e_{it} + \beta_{it} D_{it}, \quad (4)$$

where β_{it} represents a heterogeneous treatment effect, and θ_t denotes a common trend.

First, if $b_i \neq b$ for any i , [Lee and Sul \(2026\)](#) show that K_t increases over time. Second, even when $b_i = b$ for all i , K_t can increase if e_{it} is nonstationary. Consequently, [Lee and Sul \(2026\)](#) use the weak σ -convergence test to assess the parallel trends assumption.

However, weak σ -convergence is a not air tight in relation to the distribution. Even if the cross-sectional variance of y_{it} does not increase over time, this does not necessarily mean that all quantiles of y_{it} remain stable relative to the mean. Specifically, quantiles near the tails of the distribution may diverge from the center while other units converge toward the mean, resulting in an overall cross-sectional variance of y_{it} that does not diverge. Therefore, we need a more conservative definition of convergence.

Denote $K_t^\alpha = n^{-1} \sum_{i=1}^n (y_{it} - \mu_t^\alpha)^2$.

Definition 2 (Distributional σ -Convergence) *y_{it} is distributionally σ -converging (or non-diverging) if K_t^α is weakly σ -converging (or non-diverging) across all $0 < \alpha < 1$.*

If distributional σ -convergence (or non-divergence) holds, then weak σ -convergence (or non-divergence) automatically follows. However, the reverse is not true. Only when $d_{G,t}^\alpha$ for $G = N, I$ is not diverging over time, weak σ -convergence implies distributional σ -convergence. See Theorem 1 in next section for more detailed discussions.

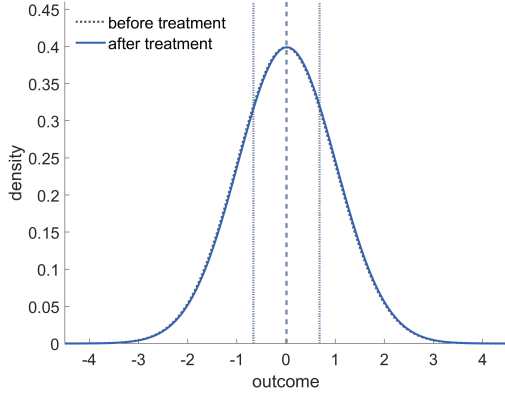
Figure 1 illustrates the relation between weak and distributional σ -convergence. Panels (a) and (c) show that when dispersion around the quantiles are stable, dispersion around the mean is also stable. Hence, distributional σ -convergence implies weak σ -convergence. Panels (b) and (d) show the converse failure: the mean-centered dispersion can remain stable while the quantile-centered dispersion rises. Weak σ -convergence can therefore miss tail-specific instability.

2.2 Assumptions

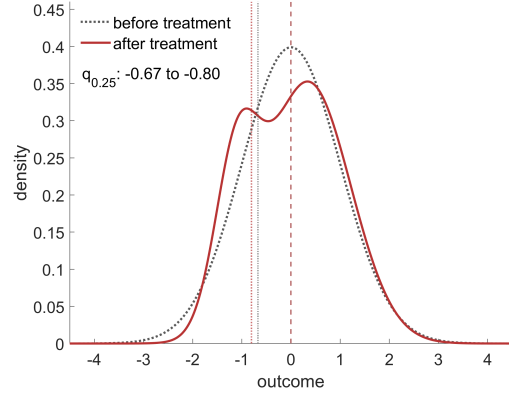
Let $\mathcal{T}_0 = \{1, \dots, T_0\}$ denote the pre-treatment period, $\mathcal{T}_1 = \{1, \dots, T_1\}$ where $T_0 < T_1 \leq T$ denote the post treatment time period of interest, and $\mathcal{T}_2 = \{T_0 + 1, \dots, T\}$ denote the post-treatment period, and . Let $\mathcal{A} \subset (0, 1)$ be a compact set of quantile indexes. For $G \in \{N, I\}$, define the following dispersion statistics.

$$\begin{aligned} K_{G,t} &= \frac{1}{n_{G,t}} \sum_{i=1}^{n_{G,t}} (y_{it}^G - \mu_{G,t})^2, \\ K_{G,t}^\alpha &= \frac{1}{n_{G,t}} \sum_{i=1}^{n_{G,t}} (y_{it}^G - \mu_{G,t}^\alpha)^2, \\ K_{N,t}^{\Delta(\alpha)} &= n_{N,t}^{-1} \sum_{i=1}^{n_N} (y_{it}^N - \mu_{I,t}^\alpha)^2 \end{aligned} \tag{5}$$

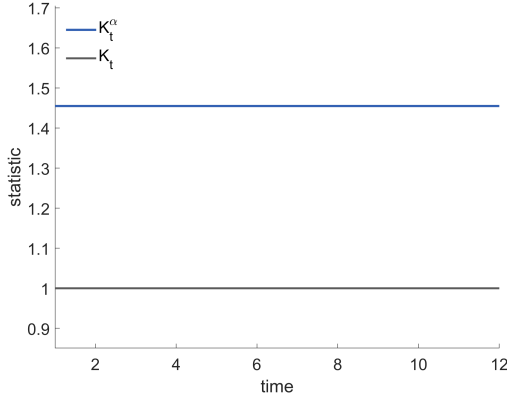
Figure 1: Weak Sigma Convergence vs Distributional Sigma Convergence



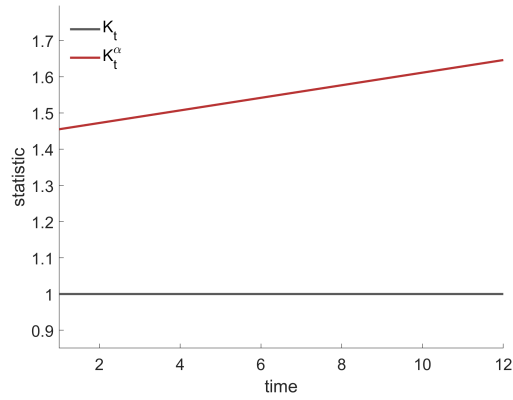
(a) Distributional σ -convergence implies weak σ -convergence



(b) Weak σ -convergence does not imply distributional σ -convergence



(c) Stable quantile dispersion also stabilizes mean dispersion



(d) Distributional statistic detects quantile instability

These three dispersion statistics play different roles. $K_{G,t}$ measures dispersion around the group mean and is used to test weak σ -divergence. $K_{G,t}^\alpha$ measures dispersion around the group α -quantile and is used to test distributional σ -divergence. The latter is needed because mean stability does not rule out changes in the shape of the distribution. Finally, $K_{N,t}^{\Delta(\alpha)}$ measures the distance between the retained control distribution and the treated α -quantile.

Assumption A1 (Mean pre-condition) *Before treatment, treated and control outcomes have the same expected mean:*

$$\mathbb{E}[y_{it}^I(0)] = \mathbb{E}[y_{it}^N(0)], \quad t \in \mathcal{T}_0. \quad (6)$$

Since treated units are untreated during $\mathcal{T}_0$, this condition is testable in the pre-treatment sample using observed outcomes. It rules out pre-treatment mean differences between the ever-treated and control groups before applying the distributional diagnostics.

Assumption A2 (Non-diverging control distribution) *The control group satisfies weak and distributional σ -convergence over the full sample. That is, $K_{N,t} < \infty$ and $K_{N,t}^\alpha < \infty$ a.s. for all t and $\alpha \in \mathcal{A}$, with $\text{plim}_{t \rightarrow \infty} K_{N,t} \in [0, \infty)$, $\text{plim}_{t \rightarrow \infty} K_{N,t}^\alpha \in [0, \infty)$ for all $\alpha \in \mathcal{A}$, and, for some increasing sequence $c_T \rightarrow \infty$,*

$$\limsup_{T \rightarrow \infty} c_T^{-1} \sum_{t=1}^T \tilde{K}_{N,t} \tilde{t} \leq 0, \quad \sup_{\alpha \in \mathcal{A}} \limsup_{T \rightarrow \infty} c_T^{-1} \sum_{t=1}^T \tilde{K}_{N,t}^\alpha \tilde{t} \leq 0 \quad a.s. \quad (7)$$

If the raw control group violates this condition, the sieving step is applied until the retained control set satisfies the same restrictions. Assumption A2 retains non-diverging control group: neither its mean-based dispersion nor its quantile-based dispersion has a positive long-run trend.

Assumption A3 (Pre-treatment distributional comparability) *Before treatment, the ever-treated group satisfies weak and distributional σ non-divergence, and its quantiles do not diverge from the control group distribution. That is, for $t \in \mathcal{T}_0$, $K_{I,t} < \infty$, $K_{I,t}^\alpha < \infty$, and $K_{N,t}^{\Delta(\alpha)} < \infty$ a.s., with $\text{plim}_{t \rightarrow \infty} K_{I,t} \in [0, \infty)$, $\text{plim}_{t \rightarrow \infty} K_{I,t}^\alpha \in [0, \infty)$, $\text{plim}_{t \rightarrow \infty} K_{N,t}^{\Delta(\alpha)} \in [0, \infty)$ for all $\alpha \in \mathcal{A}$, and, for some increasing sequence $c_{T_0} \rightarrow \infty$,*

$$\begin{aligned} \limsup_{T_0 \rightarrow \infty} c_{T_0}^{-1} \sum_{t \in \mathcal{T}_0} \tilde{K}_{I,t} \tilde{t} &\leq 0, \\ \sup_{\alpha \in \mathcal{A}} \limsup_{T_0 \rightarrow \infty} c_{T_0}^{-1} \sum_{t \in \mathcal{T}_0} \tilde{K}_{I,t}^\alpha \tilde{t} &\leq 0, \\ \sup_{\alpha \in \mathcal{A}} \limsup_{T_0 \rightarrow \infty} c_{T_0}^{-1} \sum_{t \in \mathcal{T}_0} \tilde{K}_{N,t}^{\Delta(\alpha)} \tilde{t} &\leq 0 \quad a.s. \end{aligned} \quad (8)$$

Assumption A3 rules out pre-treatment distributional drift in the treated group and in the treated-control quantile distance.

Assumption A4 (Constant average effect when ATT is nonzero) *When the post-treatment ATT is significantly different from zero, the expected ATT path is stable:*

$$\Delta_t - \Delta_{t-1} = 0, \quad t, t-1 \in \mathcal{T}_2, \quad \text{where } \Delta_t = \mathbb{E}[y_{it}^I] - \mathbb{E}[y_{it}^N]. \quad (9)$$

Equivalently, $\Delta_t = \Delta \neq 0$ over the post-treatment period. Assumption A4 is needed when the ATT is nonzero to rule out a changing mean effect, so that when the treated distribution is also weakly and distributionally σ -non-diverging, a nonzero ATT implies a common location shift rather than a time-varying distributional effect.

Assumption A5 (Bounded mean-quantile distance during distributional divergence) *When outcome for a group, y_{it}^G , have distributional divergence, $e_{G,t}$ is bounded by $O_p(1)$ in sample mean-quantile distance $\hat{d}_{G,t}^\alpha = c_\alpha t^\gamma + e_{G,t}$.*

When $\gamma = 0$, it is straight forward to show that $e_{G,t}$ is bounded by $O_p(n_G^{1/2})$. But, when y_{it}^G is not distributionally stable as indicated by $\gamma > 0$, we assume that $e_{G,t}$ is bounded by $O_p(1)$. As long as dominant term is the first term, this assumption doesn't pose any threat to the proof.

2.3 Distributional Stability and Distributional Parallel Trend

The diagnostic proposed in this paper is motivated by a simple restriction on the untreated outcome process. For a DGP in equation (4), the untreated outcome for unit i satisfies

$$y_{it} = a_i + b_i \theta_t + e_{it} \quad (10)$$

The equation for untreated outcome for units in each group $G \in \{N, I\}$ can be further written as

$$y_{it}^G = a_i^G + b_i^G \theta_t + e_{it}^G,$$

The common factor θ_t^G may be nonstationary or may contain a stochastic trend. The key restriction is not that untreated outcomes are stationary in levels, but that units within the retained group have the same exposure to the common factor. That is, $b_i^G = b^G$, a constant,

for all retained units before treatment. Then, after removing the cross-sectional mean,

$$x_{it}^G = y_{it}^G - \hat{\mu}_t^G = a_i^G - \hat{a}_N^G + e_{it}^G - \hat{e}_{N,t}^G$$

where $\hat{\mu}_t^G$, $\hat{a}_N^G$, and $\hat{e}_{N,t}^G$ are the sample cross-sectional mean of y_{it}^G , a_i^G , and e_{it}^G respectively. Thus, the common component $b_i^G \theta_t$ is absorbed by the cross-sectional mean. The quantile of x_{it}^G is n -asymptotically equivalent to that of $a_i^G + e_{it}^G$. If the distribution of e_{it}^G is stable over time, or more generally if the cross-sectional distribution of $a_i^G + e_{it}^G$ has distributional non-divergence, the retained group is distributionally stable. In this sense, distributional stability does not require the untreated outcome level to be stationary. It requires that the cross-sectional distribution around the common component does not drift over time.

The control distribution must not diverge over time, including time periods after treatment begins; otherwise, the comparison itself becomes impossible. Recall, $\hat{d}_{N,t}^\alpha = \hat{\mu}_{N,t} - \hat{\mu}_{N,t}^\alpha$, the sample analog of $d_{N,t}^\alpha$. Suppose $d_{N,t} = c_\alpha^N$ so that the distance between the true mean and the α -th quantile is time invariant. As [Chernozhukov, Fernández-Val, and Melly \(2013\)](#) show, $\hat{\mu}_{N,t}^\alpha = \mu_{N,t}^\alpha + O_p(n_N^{-1/2})$. The smaller order term is influenced by the estimated quantile of e_{it} . As long as c_α^N does not increase over time, Theorem 1 shows that weak σ -convergence implies distributional σ -convergence. Meanwhile, if c_α^N increases over time for some $\alpha > 0$, then even though y_{it} is weakly σ -converging to its cross-sectional mean, some of y_{it} diverge from the α -th quantile. We require the same condition for ever-treated group before treatment.

The control-group distributional stability condition must hold over all periods in which the control group is used as the untreated benchmark. Pre-treatment stability is not sufficient if the control distribution becomes unstable after treatment begins. For example, an unobserved policy or aggregate shock may affect some control units after the treatment date. Then some controls may load on a different factor, say θ_t^* , rather than on θ_t . In that case, the post-treatment control distribution is no longer a valid counterfactual benchmark for QTT. The procedure therefore tests the retained control group over the full sample period. If the retained control group fails the weak or distributional σ -non-divergence tests, the control group is sieved before it is used for distributional DID or QTT. In contrast, post-treatment divergence of treated-group is not automatically a failure. It is part of the treatment-effect evidence.

We define distributional parallel trends as the equality of mean-to-quantile distances in treated and control group before treatment given by equation (11). The condition says that, before treatment, the treated and control groups may differ in fixed components, but

their untreated distributions should not diverge around their means or around the relevant quantiles.

$$\mu_{I,t} - \mu_{I,t}^\alpha = \mu_{N,t} - \mu_{N,t}^\alpha, \quad t \in \mathcal{T}_0, \quad \alpha \in \mathcal{A}, \quad (11)$$

Next, $\hat{\Delta}_t^\alpha = \hat{\mu}_{I,t}^\alpha - \hat{\mu}_{N,t}^\alpha = \mu_{I,t}^\alpha - \mu_{N,t}^\alpha - (\mu_{I,t}^\alpha - \hat{\mu}_{I,t}^\alpha) + (\mu_{N,t}^\alpha - \hat{\mu}_{N,t}^\alpha) = \Delta_t^\alpha + O_p(n_I^{-1/2}) + O_p(n_N^{-1/2}) = \Delta_t^\alpha + O_p(n_{\min}^{-1/2})$ where $n_{\min} = \min\{n_I, n_N\}$. Note that $\mu_{N,t}^\alpha - \mu_{I,t}^\alpha = \mu_{N,t}^\alpha - \mu_{N,t} - \mu_{I,t}^\alpha + \mu_{N,t} = 0$ only when $\mu_{N,t} = \mu_{I,t}$ and $c_\alpha^N = c_\alpha^I$. Assumption A1 satisfies the first condition. [Roth and Sant'Anna \(2023\)](#) found that the mean parallel trends will be insensitive to functional forms if $c_\alpha^N = c_\alpha^I$. We use this result to assume that the cross-sectional variance of control group and treated group in each period before treatment is equal.

Within-group non-divergence does not by itself imply pre-treatment treated-control quantile equality. Even if $d_{N,t}^\alpha$ and $d_{I,t}^\alpha$ are constant, the pre-treatment quantile gap satisfies $\mu_{I,t}^\alpha - \mu_{N,t}^\alpha = (\mu_{I,t} - \mu_{N,t}) - (d_{I,t}^\alpha - d_{N,t}^\alpha)$. Hence the gap is zero only when the pre-treatment mean gap and the pre-treatment mean-to-quantile distances are compatible. We therefore require for distributional parallel trends, implemented by testing that $K_{N,t}^{\Delta(\alpha)}$ does not diverge over $\mathcal{T}_0$. Without this condition, the diagnostic classification for the zero-ATT case is not valid.

2.4 Distributional Non-Divergence and Quantile Treatment on the Treated

This subsection describes the relationship between the concept of distributional non-divergence and the quantile treatment on the treated. After the assumptions are satisfied, we test the distributional divergence of treated group using $\hat{K}_{I,t}^\alpha$ in trend regression equation (14) over time period $\mathcal{T}_1$. In addition, we also estimate the ATT in post treatment period. We have three cases based on the distributional divergence of treated group and the estimated ATT.

(i) After treatment, y_{it}^I also demonstrates distributional non-divergence after treatment.

Condition (i) is crucial. If it holds, the quantiles of y_{it}^N do not diverge from their sample mean, and the same applies to y_{it}^I . This indicates that the diagnostic does not detect quantile-specific instability beyond the mean effect. If this condition holds along with Assumptions A1–A3 and the estimated ATT is zero, the evidence supports the case in which there is no empirically relevant QTT. If the estimated ATT is stable and nonzero along with Assumption A4, the evidence supports a common location-shift interpretation. This scenario is demonstrated in panel (a) and (c) of figure 2. For a case with positive ATT,

the treated distribution shifts to the right relative to the control distribution, but its shape remains stable: the median and the quantile band move together. If the estimated ATT is zero, the pre-treatment distribution and quantiles would coincide with the ones in post treatment. QTT can still be estimated, but the diagnostic does not indicate that QTT is empirically necessary.

Next, we examine two additional cases where the treated group displays distributional divergence after treatment, combined with Assumptions A1–A3.

- (ii) After treatment, y_{it}^I displays distributional divergence, but the estimated ATT is not statistically significant.
- (iii) After treatment, y_{it}^I shows distributional divergence, and the estimated ATT is statistically significant.

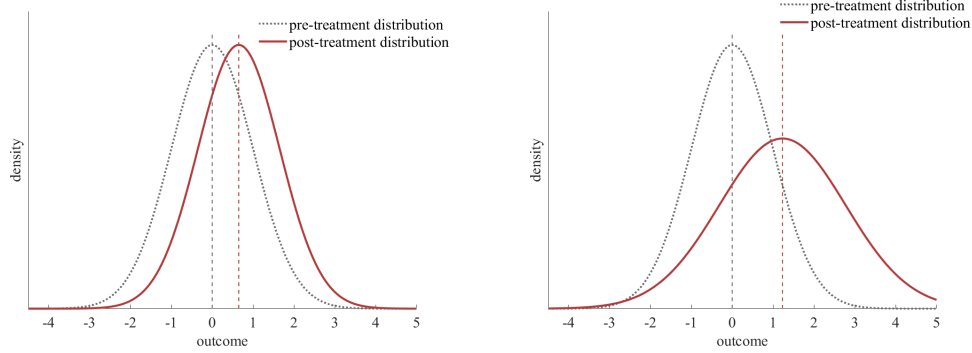
In case (ii), the ATT can be particularly misleading. Although the group mean effect is zero, either or both low and high quantiles of y_{it}^I may differ from those of y_{it}^N . Thus, estimating the QTT is essential to understand the distributional effects of the treatment.

Case (iii) presents a more complex scenario. The need to estimate the QTT arises only when the ATT is greater than or less than zero, and y_{it}^N is σ -diverging from the low (or high) quantile of y_{it}^I . If the ATT is greater than zero (or $\hat{\mu}_{N,t} < \hat{\mu}_{I,t}$), then $\hat{\mu}_{N,t}^\alpha < \hat{\mu}_{I,t}^\alpha$ for $0.5 < \alpha < 1$, with the gap between the quantiles typically widening over time. However, the distance between the quantiles will not necessarily increase when $\alpha < 0.5$. This situation occurs only when the entire distribution of y_{it}^I shifts to the right much more rapidly than the dispersion between the quantiles and the sample mean.

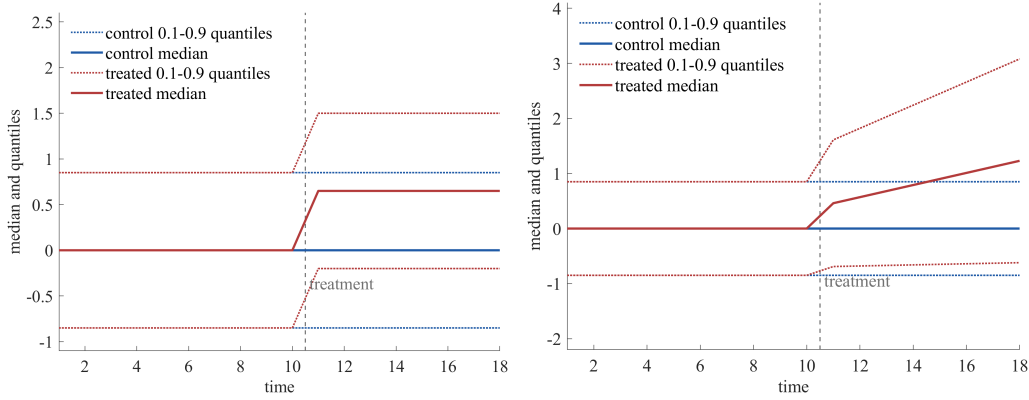
Generally, for lower quantiles, the distance between $\hat{\mu}_{N,t}^\alpha$ and $\hat{\mu}_{I,t}^\alpha$ decreases. In other words, the rate of mean shift (ATT) is slower than the rate of quantile shift. In such cases, estimating the QTT, particularly for the low quantile, is valuable for understanding the treatment's distributional effects.

Panel (b) and (c) in Figure 2 shows why the distributional diagnostic is needed even when the mean effect is known. The estimated ATT is positive and the treated distribution also diverges. The positive mean shift is real, but it does not summarize the full response because the tails move differently from the center. If the estimated ATT is zero, the entire distribution of treated group post treatment shifts to left. In both cases, QTT is needed to recover the sign and magnitude of the quantile-specific effects. The divergence statistic tells us that separation exists; direct QTT estimation tells us where the treated quantile lies relative to its control distribution.

Figure 2: Distribution of treated group after treatment



(a) No distributional σ divergence in treated group (b) Distributional σ divergence in treated group



(c) Quantile paths in non-distributional divergence case (d) Quantile paths in distributional divergence case

3 Implementations and Asymptotic Justification

Implementation of pre-test for QTT using distributional σ -convergence begins with data preparation.

3.1 Data Preparation

DID estimation requires assessing the credibility of parallel trends assumption before actual estimation performed. For distributional comparison, we need stronger restrictions. First, we review the modified parallel trends test (MPTA) proposed by [Lee and Sul \(2026\)](#). Under the data generating process (DGP) in (4), the null hypothesis of the MPTA test can be written as

$$H_0 : b_i = b \text{ for all } i. \quad (12)$$

Let $y_{it}^* = y_{it}(1 - D_{it})$. Under the assumption of stationary e_{it} , [Lee and Sul \(2026\)](#) show that weak σ -convergence of y_{it}^* from $t = 1$ to $t = T_0$ implies the MPTA. If the MPTA does not hold, they suggest sieving outlying units for which $b_i \neq b$. Let $\mathcal{G}$ denote the sieved set of units that satisfies the MPTA condition.

The ideal condition is that the distributions of control units before treatment are the same as those of ever-treated units. Hence, the Kolmogorov-Smirnov (KS) test could be used for testing the equal distribution for each $t = 1, \dots, T_0$. However, it is well known that the power of the KS test is poor. Instead, we examine whether or not the distance between a quantile of y_{it}^G and the sample mean $\mu_{G,t}$ is not increasing over time. A direct test is not desirable if T_0 is small. To increase the power of the test, we use the distributional divergence test. Consider the sample dispersion statistics corresponding to equation (5).

$$\begin{aligned} \hat{K}_{G,t} &= \frac{1}{n_{G,t}} \sum_{i=1}^{n_{G,t}} (y_{it}^G - \hat{\mu}_{G,t})^2, \text{ with } G = \{N, I\} \\ \hat{K}_{G,t}^\alpha &= \frac{1}{n_{G,t}} \sum_{i=1}^{n_{G,t}} (y_{it}^G - \hat{\mu}_{G,t}^\alpha)^2, \text{ with } G = \{N, I\} \\ \hat{K}_{N,t}^{\Delta(\alpha)} &= n_{N,t}^{-1} \sum_{i=1}^{n_N} (y_{it}^N - \hat{\mu}_{I,t}^\alpha)^2. \end{aligned} \quad (13)$$

To investigate whether y_{it}^N and y_{it}^I in the set $\mathcal{G}$ are distributionally non-diverging, we run the following trend regression with $\hat{K}_{G,t}^\alpha$ in equation (14) where $t = 1, \dots, T_0$ with $G = I$ but $t = 1, \dots, T$ with $G = N$.

$$\begin{aligned} \hat{K}_{G,t} &= \phi_{G,0} + \phi_{G,t} + u_t, \text{ with } G = \{N, I\}, \\ \hat{K}_{G,t}^\alpha &= \phi_{G,0,\alpha} + \phi_{G,\alpha}t + u_t, \text{ with } G = \{N, I\} \\ \hat{K}_{N,t}^{\Delta(\alpha)} &= \phi_{N,0,\Delta,\alpha} + \phi_{N,\Delta,\alpha}t + u_t. \end{aligned} \quad (14)$$

The null hypothesis for the t-statistics of trend coefficients these regressions is given in equation (15).

$$\begin{aligned} H_{0,1} &: t_{\hat{\phi}_G} \leq 1.65 \text{ for all } 0 < \alpha < 1, \\ H_{0,2} &: t_{\hat{\phi}_{G,\alpha}} \leq 1.65 \text{ for all } 0 < \alpha < 1, \\ H_{0,3} &: t_{\hat{\phi}_{N,\Delta,\alpha}} \leq 1.65 \text{ for all } 0 < \alpha < 1. \end{aligned} \quad (15)$$

$\hat{K}_{G,t}^\alpha$ is the cross-sectional average of the squared difference between outcomes and the α -th quantile. Similar to (2), construct the t -ratio for $\hat{\phi}_{G,\alpha}$ with $G = \{N, I\}$, denoted by $t_{\hat{\phi}_{G,\alpha}}$. The null hypothesis of no distributional divergence is given by $H_{0,1}$ in equation (15).

If $t_{\hat{\phi}_{G,\alpha}} > 1.65$, then the null hypothesis is rejected and y_{it}^G is distributionally diverging over time, which means that either lower or upper quantile is diverging from its cross-sectional average over time.

Next, we recommend examining the distributional divergence with $\hat{K}_{G,t}^{\Delta(\alpha)}$. If y_{it}^N does not diverge from any $\hat{\mu}_{I,t}^\alpha$, then we can conclude that the distribution of y_{it}^N is changing in parallel with that of y_{it}^I before treatment. For this, we run the trend regression with $\hat{K}_{N,t}^{\Delta(\alpha)}$ as specified in (14). Similar to (2), the t -ratio for $\hat{\phi}_{N,\Delta\alpha}$ must be constructed. The null hypothesis of no distributional divergence is given $H_{0,3}$ in equation (15). If $t_{\hat{\phi}_{N,\Delta,\alpha}} > 1.65$, then the null hypothesis is rejected and y_{it}^N is distributionally diverging from $\hat{\mu}_{I,t}^\alpha$ over time.

If $y_{it} \in \mathcal{G}$ exhibits distributional divergence, divergent outliers should be sieved out. Otherwise, the control distribution is not stable over time, the estimated quantiles are unstable, and the quantile comparison between treated and control units becomes unreliable. In the technical appendix, we provide a modified sieving algorithm based on Lee and Sul (2026).

3.2 Asymptotic Justification

We provide the asymptotic justification of the implementation procedures discussed in the previous subsection. For a given dispersion statistics around the quantile, $\hat{K}_{G,t}^\alpha$ can be rewritten as

$$\hat{K}_{G,t}^\alpha = \frac{1}{n_G} \sum_{i=1}^n (y_{it}^G - \hat{\mu}_{G,t}^\alpha)^2 = \hat{K}_{G,t} + (\hat{d}_{G,t}^\alpha)^2,$$

where $\hat{d}_{G,t}^\alpha = \hat{\mu}_{G,t} - \hat{\mu}_{G,t}^\alpha = c_\alpha t^\gamma + e_{G,t}$. There are two cases based on the value of γ .

The first case corresponds to when $\gamma = 0$ showing a stable mean-to-quantile distance within group G as $e_{G,t}$ is bounded by $O_p(n_G^{-1/2})$. This arises when the distribution of y_{it}^G does not spread out over time, so that $\mathbb{E}[\hat{d}_{G,t}^\alpha] = c_\alpha$. In this case, the quantile-centered statistic $\hat{K}_{G,t}^\alpha$ differs from $\hat{K}_{G,t}$ only by a constant term, which is absorbed by the intercept.

The second case corresponds to when $\gamma > 0$ implying growing within-group distributional dispersion and $e_{G,t}$ is bounded by $O_p(1)$. When $\mathbb{E}[\hat{d}_{G,t}^\alpha] = c_\alpha t^\gamma$, the distance between the group mean and the α -quantile grows over time. Then $\hat{K}_{G,t}^\alpha$ inherits a positive trend, and the slope test rejects distributional non-divergence.

Now we are ready to present the core theorem.

Theorem 3 (Asymptotic Limits for Within Group Distributional σ - Divergence)

As $n, T \rightarrow \infty$ and $T/n \rightarrow 0$,

(i) if $\hat{d}_{G,t}^\alpha = c_\alpha + O_p(n_G^{-1/2})$, then the group doesn't exhibit distributional σ -divergence as

$$t_{\hat{\phi}_{G,\alpha}} \xrightarrow{d} \mathcal{N}(0, 1),$$

(ii) If $\hat{d}_{G,t}^\alpha = c_\alpha t^\gamma + O_p(1)$ with c_α and $\gamma > 0$, then the group is distributionally σ -diverging as

$$t_{\hat{\phi}_{G,\alpha}} \rightarrow +\infty.$$

Theorem 1 is the distinction between weak σ -convergence and distributional σ -convergence. When the distance between the mean and the α -quantile is constant, the quantile-centered statistic is only a vertical shift of the dispersion statistic, and the two diagnostics have the same asymptotic content. However, when the mean-to-quantile distance grows like $c_\alpha t^\gamma$, the quantile-centered statistic contains a positive deterministic trend even if the dispersion statistic itself is non-diverging.

Remark 4 (Constant versus non-increasing mean-to-quantile distance) *The required condition is not literally that $\hat{d}_{G,t}^\alpha$ must be constant at every date. What is needed for the slope test is that $(\hat{d}_{G,t}^\alpha)^2$ has no positive trend. Thus $\hat{d}_{G,t}^\alpha$ may fluctuate over time as long as it remains within a tight boundary and does not generate an increasing deterministic component. To avoid potential confusion, the theorem states the clean constant case, $\hat{d}_{G,t}^\alpha = c_\alpha + O_p(n_G^{-1/2})$. This is the empirically useful benchmark: if the mean-to-quantile distance is non-divergent, then weak σ convergence implies distributional σ -convergence.*

Remark 5 (Within group pre-treatment case) *For the pre-treatment period $\mathcal{T}_0$, we apply these tests separately to the control group and the ever-treated group. The goal is to verify that each group is non-divergent both around its mean and around its quantiles before treatment begins. If either group exhibits divergence, we apply the sieving algorithm described in the Appendix to remove divergent units and construct retained groups that satisfy the required non-divergence conditions.*

Remark 6 (Within group post-treatment case) *For the post-treatment period $\mathcal{T}_1$, we apply the test to the control group to verify that the control distribution remains non-divergent.*

If the control group exhibits distributional divergence, we use the sieving algorithm described in the Appendix to construct a retained non-divergent control group. For the treated group, however, the post-treatment test is used only as a diagnostic for classification. We do not sieve treated units after treatment, because post-treatment divergence in the treated group is part of the treatment-effect evidence.

Next, we consider cases for cross-group convergence and divergence. Note that $\hat{K}_{N,t}^{\Delta(\alpha)}$ can be rewritten as

$$\hat{K}_{N,t}^{\Delta(\alpha)} = \frac{1}{n} \sum_{i=1}^n (y_{it}^N - \hat{\mu}_{I,t}^\alpha)^2 = \hat{K}_{N,t}^\alpha + (\hat{\Delta}_t^\alpha)^2 - 2\hat{\Delta}_t^\alpha e_t,$$

where $\hat{\Delta}_t^\alpha = \hat{\mu}_{I,t}^\alpha - \hat{\mu}_{N,t}^\alpha = c_\alpha^\Delta t^\gamma + e_{N,t}^\Delta$. There are two cases based on the value of γ .

The first case corresponds to a stable quantile gap when $\gamma = 0$ when $e_{N,t}^\Delta$ is bounded by $O_p(n_{min}^{-1/2})$. In the treatment effect model (4), this arises when treatment shifts the treated distribution by a time-invariant component, so that $\beta_{it} = \beta$. In that case, the treated-control quantile distance $\mathbb{E}[\hat{\Delta}_t^\alpha]$ is constant over time and the term $(\hat{\Delta}_t^\alpha)^2$ is absorbed by the intercept of the trend regression.

The second case corresponds to a growing quantile gap arising when $\gamma > 0$ and $e_{N,t}^\Delta$ is bounded by $O_p(1)$. This arises when the treatment effect contains a time-varying component, so that $\beta_{it} = \beta_t$, or more generally when treatment heterogeneity grows over time. Then $\hat{\Delta}_t^\alpha$ follows a trend, $\hat{\Delta}_t^\alpha = c_\alpha^\Delta t^\gamma + O_p(1)$, and the squared-distance statistic $\hat{K}_{N,t}^{\Delta(\alpha)}$ inherits a positive trend. Thus the slope test rejects because the treated quantile is moving away from the control distribution.

Theorem 7 (Asymptotic Limits for Cross-Group Divergence)

As $n, T \rightarrow \infty$ and $T/n \rightarrow 0$,

(i) if $\hat{\Delta}_t^\alpha = c_\alpha^\Delta + O_p(n_{min}^{-1/2})$, then there is no cross-group distributional divergence

$$t_{\hat{\phi}_{N,\Delta,\alpha}} \xrightarrow{d} \mathcal{N}(0, 1).$$

(ii) If $\hat{\Delta}_t^\alpha = c_\alpha^\Delta t^\gamma + O_p(1)$ with c_α^Δ and $\gamma > 0$, then the group is distributionally σ -diverging

as

$$t_{\hat{\phi}_{N,\Delta,\alpha}} \rightarrow +\infty.$$

Theorem 2 gives the same logic for the cross group statistics that Theorem 1 gives for the within-group statistic. Once the control group is not distributionally σ diverging, a positive trend in $\hat{K}_{N,t}^{\Delta(\alpha)}$ is evidence that the treated α -quantile is moving away from the control distribution. The statistic is deliberately based on squared distance, so it detects separation but not sign. A rejection tells the researcher that QTT should be estimated at the flagged quantile to determine direction and magnitude.

Remark 8 (Cross group pre-treatment case) *For the pre-treatment period $\mathcal{T}_0$, we apply this test to the cross-group distance between the control distribution and the treated-group quantiles. The purpose is to verify that treated-control quantile distances do not diverge before treatment begins. This pre-treatment check ensures that the treated and control groups remain distributionally comparable prior to treatment.*

Remark 9 (Cross group post-treatment case) *For the post-treatment period $\mathcal{T}_1$, we apply the cross-group distance test to examine whether treated-group quantiles move away from the retained control distribution after treatment begins. This test is used as a diagnostic for treatment-effect classification. If the statistic diverges at a given quantile, the evidence indicates treated-control quantile separation at that part of the distribution. Because the statistic is based on squared distances, rejection indicates separation but not the sign or magnitude of QTT. Direct QTT estimation is therefore required for the affected quantiles.*

3.3 Performing Distributional Divergence Test After Treatment

The post-treatment tests therefore ask two related questions. First, does the treated distribution become more dispersed around its own mean and its quantiles? Second, do treated quantiles separate from the non-diverging control groups? The first questions concern the stability of the treated distribution. The second question concerns the treated-control comparison that is directly relevant for QTT.

Let $\mathcal{T}_1 = \{1, \dots, T_1\}$, where $T_1 > T_0$ is the time period after treatment in which the researcher wants to determine if QTT is required or not. We organize the post-treatment evidence into two diagnostic blocks.

1. Treated group weak and distributional σ convergence

First, we examine whether the treated group becomes more dispersed around its own mean or around selected treated quantiles after treatment. For the mean-centered diagnostic, we compute the treated-group cross-sectional variance $\hat{K}_{I,t}$ in equation (13). To investigate whether y_{it}^I exhibits σ -divergence, we run the trend regression in equation (14) with $\hat{K}_{I,t}$. The t -ratio for $\hat{\phi}_I$ constructed as in (2) determines whether the treated group diverges using the null of (15).

This test asks whether the treated distribution spreads out around its own mean after treatment. A rejection means that treated units are becoming more dispersed around the treated mean. This warns that the ATT may be incomplete, but it does not reveal which part of the treated distribution is changing.

For the distributional divergence test in treated group, compute for dispersion statistics $\hat{K}_{I,t}^\alpha$ using (13) and run the trend regression in (14) for selected quantiles α . The rejection of null hypothesis $H_{0,2}$ for $t_{\hat{\phi}_{I,\alpha}}$ indicates the treated group exhibits distributional divergence.

For $t \in \mathcal{T}_1$ and $\alpha \in \mathcal{A}$, define $\widehat{QTT}_t^\alpha = \hat{\mu}_{I,t}^\alpha - \hat{\mu}_{N,t}^\alpha$, $\widehat{ATT}_t = \hat{\Delta}_t = \hat{\mu}_{I,t} - \hat{\mu}_{N,t}$.

Corollary 10 (Case (i): treated distributional non-divergence) *If the post-treatment treated group satisfies weak and distributional non-divergence, i.e. $t_{\hat{\phi}_I} \leq 1.65$ and $\sup_{\alpha \in \mathcal{A}} t_{\hat{\phi}_{I,\alpha}} \leq 1.65$, then the diagnostics do not reject quantile-specific divergence in the treated group. Under the pre-treatment quantile-equality restriction and the non-divergence conditions in Assumptions A1–A4, this implies*

$$\widehat{QTT}_t^\alpha = \begin{cases} O_p(n_{\min}^{-1/2}), & \text{when } \mathbb{E}[\widehat{ATT}_t] = \Delta_t = 0, \\ \Delta_t + O_p(n_{\min}^{-1/2}), & \text{when } \mathbb{E}[\widehat{ATT}_t] = \Delta_t \neq 0. \end{cases}$$

where $n_{\min} = \min\{n_I, n_N\}$.

Thus, case (i) is the diagnostic case in which the ATT is sufficient: zero ATT implies zero QTT, while a constant nonzero ATT implies a common location shift.

The quantile-centered test complements $\hat{K}_{I,t}$ because aggregate variance can hide tail-specific movement. A rejection at a lower or upper quantile means that the corresponding part of the treated distribution is becoming unstable even if the mean-centered statistic is

quiet. When both treated-side diagnostics fail to reject, the treated group is classified as distributionally non-divergent after treatment.

Economically, this is the case in which treatment may shift the average outcome, but the diagnostics do not reveal a change in the shape of the treated distribution. The ATT is therefore the natural summary object. If any treated group test rejects, Corollary 10 no longer applies and QTT is required for determining the distributional effect.

2. Treated-control quantile separation.

We compute the cross-group quantile-separation statistic $\hat{K}_{N,t}^{\Delta(\alpha)}$ using equation (13) over $\mathcal{T}_1$ for further diagnostics. This statistic uses the non-distributionally diverging control distribution as the benchmark and asks whether the treated α -quantile is moving away from the control distribution after treatment. We perform this test using the trend regression in (14) with $\hat{K}_{N,t}^{\Delta(\alpha)}$ over $\mathcal{T}_1$. The null is that the squared distance from control outcomes to the treated α -quantile does not have a positive trend, as $H_{0,3}$ in (15). If $t_{\hat{\phi}_{N,\Delta,\alpha}} > 1.65$, then the treated α -quantile is dynamically separating from the control distribution.

Corollary 11 (Cases (ii)–(iii): cross-quantile separation) *Suppose Assumptions A1–A4 hold. For some $\alpha \in \mathcal{A}$, suppose the cross-quantile separation test rejects, $t_{\hat{\phi}_{N,\Delta,\alpha}} > 1.65$. Then the estimated quantile effect at α is not summarized by the estimated ATT.*

$$\widehat{QTT}_t^\alpha = \begin{cases} c_\alpha^\Delta t^\gamma + \epsilon_t + O_p(n_{\min}^{-1/2}), & \text{when } \mathbb{E}[\widehat{ATT}_t] = \Delta_t = 0, \\ \Delta_t + c_\alpha^\Delta t^\gamma + \epsilon_t + O_p(n_{\min}^{-1/2}), & \text{when } \mathbb{E}[\widehat{ATT}_t] = \Delta_t \neq 0. \end{cases}$$

where $n_{\min} = \min\{n_I, n_N\}$.

In particular, if $ATT_t = 0$, so that $\widehat{ATT}_t = O_p(n_{\min}^{-1/2})$, then $\widehat{QTT}_t^\alpha = c_\alpha^\Delta t^\gamma + \epsilon_t + O_p(n_{\min}^{-1/2})$, and therefore $\widehat{QTT}_t^\alpha \neq O_p(n_{\min}^{-1/2})$, which is case (ii). If $ATT_t = \Delta \neq 0$ and $\Delta_t - \Delta_{t-1} = 0$ for $t, t-1 \in \mathcal{T}_1$, then $\widehat{ATT}_t = \Delta + O_p(n_{\min}^{-1/2})$, and $\widehat{QTT}_t^\alpha - \widehat{ATT}_t = c_\alpha^\Delta t^\gamma + \epsilon_t + O_p(n_{\min}^{-1/2})$. Thus $\widehat{QTT}_t^\alpha - \widehat{ATT}_t \neq O_p(n_{\min}^{-1/2})$, which is case (iii). Therefore QTT should be estimated at every quantile for which $t_{\hat{\phi}_{N,\Delta,\alpha}} > 1.65$. Since $\hat{K}_{N,t}^{\Delta(\alpha)}$ is a squared-distance statistic, the rejection indicates cross-quantile separation, not the sign or magnitude of the quantile effect.

Corollary 11 is a separation result, not a signed QTT estimator. Since $\hat{K}_{N,t}^{\Delta(\alpha)}$ is based on squared distances, rejection identifies the need for QTT at the flagged quantile but does not determine the sign of $\mu_{I,t}^\alpha - \mu_{N,t}^\alpha$.

In practice, the researcher does not need to test a very fine grid of quantiles. With small n_N or n_I , for example less than 200, one can select $\alpha = (0.25, 0.75)$. With larger groups, for example greater than 500, one can use $\alpha = (0.10, 0.90)$ to examine the tails. Each post-treatment test should report the estimated slope, the HAC t -statistic, the one-sided rejection rule, and the interpretation of the rejection: treated instability, quantile-specific treated instability, or treated-control quantile separation.

4 Monte Carlo Simulations

We investigate the finite-sample behavior of the classification procedure using two non-staggered Monte Carlo designs with unit fixed effects and a common time effect. The first design represents the case (i), where treatment changes only the location of the treated distribution and therefore does not create post-treatment dispersion. The second design represents the cases (ii) and (iii), where treatment generates time-growing heterogeneity among treated units and therefore creates distributional divergence. In both designs, the untreated potential outcome is

$$\begin{aligned} Y_{it}(0) &= a_i + \theta_t + \varepsilon_{it}, \\ \theta_t &= \frac{t-1}{T-1} + W_t, \quad W_t = \sum_{s=1}^t \xi_s, \quad a_i \stackrel{iid}{\sim} N(0, 1), \\ \varepsilon_{it} &= t^{-0.1} \eta_{it} + u_{it}, \quad \xi_t, \eta_{it}, u_{it} \stackrel{iid}{\sim} N(0, 1). \end{aligned} \tag{16}$$

The unit effect a_i generates persistent cross-sectional heterogeneity, while θ_t contains both a deterministic linear trend and a random-walk common component. Since θ_t is common to all units, it changes the common location of the panel but does not mechanically create cross-sectional dispersion. The idiosyncratic component combines a mildly shrinking shock, $t^{-0.1} \eta_{it}$, with a stationary shock, u_{it} . This design therefore lets the simulation ask whether the classification statistics detect genuine changes in treated dispersion and quantile separation rather than a common aggregate trend or a common random-walk shock.

For case (i), post-treatment treated observations are generated as

$$Y_{it} = Y_{it}(0) + \beta D_{it}, \quad \beta \in \{0, 0.50\}. \tag{17}$$

The treatment effect is therefore a constant location shift. When $\beta = 0$, the design corresponds to case (i) when there is no mean effect and no quantile effect. When $\beta = 0.50$, the design corresponds to case (i) when the ATT is stable and positive, and because the

treated distribution is only shifted, the quantile treatment effects inherit the same stable location change.

For cases (ii) and (iii), post-treatment treated observations are generated as

$$\begin{aligned} Y_{it} &= Y_{it}(0) + \beta_{it}D_{it}, \\ \beta_{it} &= \lambda_i t^\gamma + v_{it} + \beta, \\ \lambda_i &\sim U[-1, 1], \end{aligned} \tag{18}$$

where $v_{it} \stackrel{iid}{\sim} N(0, 1)$, $\beta \in \{0, 0.50\}$, and $\gamma \in \{0.40, 0.50\}$.

The heterogeneous component $\lambda_i t^\gamma$ is centered in expectation, but its cross-sectional spread grows with calendar time. The parameter γ controls the strength of this divergence. Smaller values create weaker quantile separation, while larger values move the design closer to the strong-divergence benchmark. The constant component β shifts the average treatment effect across the same two values used in case (i). Thus the design for cases (ii) and (iii) covers the zero-ATT case and the stable positive-ATT case while preserving the distributional-divergence channel. When $\beta = 0$, the design is the Monte Carlo analogue of case (ii), an average-effect analysis can look uninformative even though the treated distribution is separating. When $\beta = 0.50$, the design is the analogue of case (iii), the ATT is shifted by a stable common component, but the growing heterogeneity prevents the ATT from summarizing quantile effects.

Treatment timing is non-staggered. A total of 60 percent of units belong to the treated group, while the remaining 40 percent remain controls. Let $T_0 = T/2$. All treated units become active at $T_0 + 1$, and treatment remains active for all post-treatment periods. There is therefore no changing treated composition after treatment; any positive slope in the diagnostic statistics is generated by the outcome process rather than by staggered adoption.

The simulations use $R = 1000$ replications and quantile indexes $\alpha \in \{0.25, 0.50, 0.75\}$. The design varies $N \in \{100, 300, 500, 1000\}$ and $T \in \{20, 50, 100\}$. In each replication, the slope statistics are computed from the post-treatment time-series regressions of the relevant dispersion statistic on an intercept and a linear time trend. The one-sided upward-trend critical value is 1.65, and the Newey–West lag length is set to $\text{round}(T_{\text{post}}^{1/3})$.

For case (i), the simulation records the rejection behavior of $\hat{K}_{I,t}$ and $\hat{K}_{I,t}^\alpha$. Case (i) classification rate is the share of replications in which no treated quantile-dispersion statistic rejects across the three quantiles. For cases (ii) and (iii), the simulation records the rejection behavior of $\hat{K}_{I,t}$ and $\hat{K}_{N,t}^{\Delta(\alpha)}$. The cases (ii) and (iii) classification rate is the share of replications in which the treated dispersion statistic rejects and at least one treated-control

quantile-separation statistic rejects. Table 2 reports the case (i) rejection and classification rates. Table 3 reports the cases (ii) and (iii) rates for the stronger divergence design, $\gamma = 0.50$; the weaker design, $\gamma = 0.40$, is discussed in the text.

Table 1: Monte Carlo Classification Rates for Case (i)

		ATT = 0.00					ATT = 0.50				
N	T	$K_{I,t}$	Class.	$K_{I,t}^{.25}$	$K_{I,t}^{.50}$	$K_{I,t}^{.75}$	$K_{I,t}$	Class.	$K_{I,t}^{.25}$	$K_{I,t}^{.50}$	$K_{I,t}^{.75}$
100	20	0.087	0.848	0.102	0.092	0.093	0.103	0.823	0.113	0.103	0.102
100	50	0.059	0.898	0.068	0.058	0.061	0.061	0.892	0.067	0.061	0.071
100	100	0.051	0.923	0.054	0.048	0.038	0.053	0.908	0.059	0.052	0.050
300	20	0.091	0.849	0.096	0.089	0.100	0.093	0.860	0.091	0.093	0.091
300	50	0.046	0.904	0.067	0.045	0.054	0.042	0.910	0.050	0.042	0.055
300	100	0.028	0.943	0.031	0.029	0.038	0.028	0.943	0.032	0.025	0.033
500	20	0.074	0.864	0.084	0.077	0.082	0.094	0.857	0.088	0.093	0.100
500	50	0.044	0.919	0.048	0.044	0.050	0.039	0.931	0.048	0.043	0.038
500	100	0.017	0.955	0.025	0.020	0.025	0.025	0.948	0.027	0.026	0.035
1000	20	0.048	0.909	0.062	0.047	0.049	0.059	0.896	0.063	0.060	0.058
1000	50	0.020	0.954	0.024	0.021	0.028	0.027	0.947	0.031	0.027	0.022
1000	100	0.010	0.976	0.013	0.011	0.010	0.011	0.975	0.014	0.011	0.014

Notes: The table reports rejection and classification frequencies from $R = 1000$ non-staggered Monte Carlo replications for case (i). Treated outcomes equal $Y_{it}(0) + \beta D_{it}$ with $\beta \in \{0, 0.50\}$, so treatment is a pure location shift. $K_{I,t}$ is the one-sided rejection rate for treated mean-centered dispersion. $K_{I,t}^{.25}$, $K_{I,t}^{.50}$, and $K_{I,t}^{.75}$ are rejection rates for treated quantile-centered dispersion, $K_{I,t}^\alpha$, at $\alpha = 0.25, 0.50, 0.75$. Class. is the share of replications classified as case (i), meaning that none of the treated quantile-dispersion tests rejects. The critical value is 1.65 and HAC standard errors use Newey–West lag length $\text{round}(T_{\text{post}}^{1/3})$.

The case (i) simulations behave as the theory predicts. The weak and distributional σ -non-divergence classification rate is high for the zero and positive-ATT designs, while the one-sided rejection rates for $\hat{K}_{I,t}$ and $\hat{K}_{I,t}^\alpha$ remain low. For example, when $T = 100$, the case (i) classification rate is above 0.90 for every reported value of N and both reported values of β . The rejection rates for the treated mean-dispersion statistic and the treated quantile-dispersion statistics also decline as the post-treatment time dimension grows. This is the setting in which ATT is a distributionally meaningful summary: a zero ATT indicates no quantile treatment gap, while a stable positive ATT shifts quantiles by the same amount as the mean.

The cases (ii) and (iii) simulations show the opposite pattern, but the strength of the signal varies with γ . For the weaker design with $\gamma = 0.40$ (not tabulated), classification is moderate in short panels and smaller samples but rises quickly as either N or T increases. Table 3 reports the stronger design with $\gamma = 0.50$, where power is much higher: the treated dispersion statistic $\hat{K}_{I,t}$ is close to one throughout most of the grid, and the cases

Table 3: Monte Carlo Classification Rates for Cases (ii) and (iii), $\gamma = 0.50$

		ATT = 0.00					ATT = 0.50				
N	T	$K_{I,t}$	Class.	$K_{N,t}^{\Delta(.25)}$	$K_{N,t}^{\Delta(.50)}$	$K_{N,t}^{\Delta(.75)}$	$K_{I,t}$	Class.	$K_{N,t}^{\Delta(.25)}$	$K_{N,t}^{\Delta(.50)}$	$K_{N,t}^{\Delta(.75)}$
100	20	0.857	0.635	0.411	0.118	0.409	0.832	0.607	0.373	0.126	0.410
100	50	1.000	0.988	0.819	0.143	0.817	1.000	0.987	0.829	0.157	0.816
100	100	1.000	1.000	0.977	0.268	0.974	1.000	1.000	0.976	0.291	0.977
300	20	0.994	0.910	0.632	0.095	0.653	0.993	0.890	0.600	0.106	0.627
300	50	1.000	1.000	0.987	0.124	0.987	1.000	0.999	0.983	0.132	0.990
300	100	1.000	1.000	1.000	0.253	1.000	1.000	1.000	1.000	0.203	1.000
500	20	0.998	0.961	0.778	0.096	0.727	1.000	0.960	0.745	0.110	0.748
500	50	1.000	1.000	1.000	0.099	1.000	1.000	1.000	1.000	0.109	1.000
500	100	1.000	1.000	1.000	0.197	1.000	1.000	1.000	1.000	0.176	1.000
1000	20	1.000	0.997	0.895	0.071	0.910	1.000	0.997	0.903	0.089	0.913
1000	50	1.000	1.000	1.000	0.054	1.000	1.000	1.000	1.000	0.088	1.000
1000	100	1.000	1.000	1.000	0.142	1.000	1.000	1.000	1.000	0.141	1.000

Notes: The table reports rejection and classification frequencies from $R = 1000$ non-staggered Monte Carlo replications for cases (ii) and (iii) under the stronger divergence design, $\gamma = 0.50$. Treated outcomes equal $Y_{it}(0) + \beta_{it}D_{it}$, where $\beta_{it} = \lambda_i t^\gamma + v_{it} + \beta$, $\lambda_i \sim U[-1, 1]$, $v_{it} \sim N(0, 1)$, and $\beta \in \{0, 0.50\}$. $K_{I,t}$ is the treated mean-dispersion rejection rate. $K_{N,t}^{\Delta(.25)}$, $K_{N,t}^{\Delta(.50)}$, and $K_{N,t}^{\Delta(.75)}$ are treated-control quantile-separation rejection rates for $K_{N,t}^{\Delta(\alpha)}$. Class. is the share of replications in which $K_{I,t}$ rejects and at least one treated-control quantile-separation statistic rejects. The critical value is 1.65 and HAC standard errors use Newey–West lag length $\text{round}(T_{\text{post}}^{1/3})$.

(ii) and (iii) classification rate approaches one for larger samples and longer panels. The tail quantile-separation statistics, $\hat{K}_{N,t}^{\Delta(0.25)}$ and $\hat{K}_{N,t}^{\Delta(0.75)}$, carry most of the power, while the median statistic rejects less often because the simulated heterogeneity expands the tails more strongly than it moves the center of the distribution. This is the point of the classification: the diagnostic does not merely say that the ATT exists or fails to exist; it shows whether the average effect fails as a distributional summary, how strongly that failure appears, and where it appears in the outcome distribution.

5 Empirical Example: Seguro Popular and Small-Firm Employment

Bosch and Campos-Vazquez (2014) study Seguro Popular, Mexico’s expansion of non-contributory public health insurance. The policy made health insurance available outside formal employment. In a labor market with a large informal sector, this reform could weaken the link between formal jobs and access to health benefits. The paper asks whether welfare gains from expanded health insurance came at the cost of reduced formality. Its main empirical result is that total formal employment changes little on average, while employment in small and medium firms responds more negatively. This makes small-firm employment a natural setting for illustrating the diagnostic, because a near-zero average response may still hide

distributional movement across municipalities.

The exercise in this section is diagnostic only. We do not estimate signed QTT for Seguro Popular. Instead, we ask whether the control distribution is a weakly and distributionally σ -non-diverging benchmark and whether the post-treatment diagnostics indicate quantile separation that would require direct QTT estimation. Thus the empirical section should be read as an application of the pre-estimation workflow, not as a full distributional treatment-effect estimation exercise.

The empirical implementation follows the workflow in Section 5. First, we check whether the pre-treatment control group is weakly and distributionally σ -non-diverging. Second, we verify that the control benchmark passes the all-period non-divergence sieve. Third, we test post-treatment treated-group divergence and cross-group quantile separation. Finally, because the ATT is zero, we classify the application using the zero-ATT column of the four-case framework. Table 4 reports the main diagnostic objects.

The application is classified as case (ii). The pre-treatment checks pass: none of the precondition slope tests has a significantly positive trend, and the pre-treatment sieve drops no treated or control units. The control benchmark also passes the all-period non-divergence sieve, with all 389 controls retained. Thus the post-treatment evidence is not being driven by a diverging comparison group. The control statistic $K_{N,t}$ has does not reject non-divergence, while the treated statistic $K_{I,t}$ has a t -statistic of 3.16 rejecting non-divergence.

The estimated ATT path is mostly close to zero, so the mean-based conclusion is that Seguro Popular has little average effect on municipal small-firm employment. Taken alone, this would make the application look like a no-effect case. The distributional diagnostics in Table 4, however, indicate lower-tail separation from the distributionally non-diverging control benchmark.

The cross-group quantile diagnostics identify where the distributional movement appears. The 0.10 and 0.25 quantile-separation statistics have positive and statistically significant post-treatment slopes, with t -statistics of 4.97 and 4.12. The median and upper-quantile statistics do not show positive post-treatment divergence. The distributional evidence is therefore concentrated in the lower part of the municipal small-firm employment distribution.

Because the ATT is zero, an average-effect analysis alone would suggest that Seguro Popular had no effect on this outcome. The diagnostic changes the next step of the analysis. The treated distribution diverges and lower quantile gaps move away from the distributionally non-diverging control group. This is the case (ii) diagnostic case: no average effect, but evidence of lower-tail separation that ATT cannot summarize. Since the cross-quantile statistic is based on squared distances, this evidence does not determine whether the treated lower

Table 4: Seguro Popular Diagnostic Results

Panel	Object	$t_{\hat{\phi}}$	Decision
<i>A. Non-divergence sieve</i>			
Pre-treatment	Combined treated/control units	–	1392 kept, 0 dropped
All periods	Control units	–	389 kept, 0 dropped
<i>B. Pre-treatment precondition tests</i>			
Precondition	$\hat{K}_{N,t}^{\Delta(0.10)}$	-2.87	Pass
Precondition	$\hat{K}_{N,t}^{\Delta(0.25)}$	-6.96	Pass
Precondition	$\hat{K}_{N,t}^{\Delta(0.50)}$	-1.83	Pass
Precondition	$\hat{K}_{N,t}^{\Delta(0.75)}$	-1.27	Pass
Precondition	$\hat{K}_{N,t}^{\Delta(0.90)}$	-1.70	Pass
Precondition	$\hat{K}_{N,t}$	-1.97	Pass
Precondition	$\hat{K}_{I,t}$	-3.92	Pass
<i>C. Post-treatment divergence tests</i>			
Post-treatment	$\hat{K}_{N,t}$	-4.77	Non-divergent
Post-treatment	$\hat{K}_{I,t}$	3.16	Diverges
Post-treatment	$\hat{K}_{N,t}^{\Delta(0.10)}$	4.97	Diverges
Post-treatment	$\hat{K}_{N,t}^{\Delta(0.25)}$	4.12	Diverges
Post-treatment	$\hat{K}_{N,t}^{\Delta(0.50)}$	-5.05	Non-divergent
Post-treatment	$\hat{K}_{N,t}^{\Delta(0.75)}$	-5.89	Non-divergent
Post-treatment	$\hat{K}_{N,t}^{\Delta(0.90)}$	-6.23	Non-divergent
ATT condition		ATT = 0	Classification: case (ii)

Notes: The table reports the diagnostic implementation for the Seguro Popular application. Panel A reports the non-divergence sieve: the pre-treatment row applies the sieve to the combined treated and control sample before treatment, while the all-period row applies the sieve to the control group used as the benchmark. Panel B reports pre-treatment precondition tests for the treated-control quantile-distance statistics, $\hat{K}_{N,t}^{\Delta(\alpha)}$, and the mean-centered dispersion statistics, $\hat{K}_{N,t}$ and $\hat{K}_{I,t}$. Panel C reports post-treatment divergence tests. The reported t -statistics use HAC standard errors from the time-series regression of each statistic on an intercept and a linear time trend. A positive one-sided rejection is reported as Diverges; otherwise the statistic is reported as Pass or Non-divergent. Because the estimated ATT is approximately zero, rejection of $\hat{K}_{I,t}$ together with lower-tail rejection of $\hat{K}_{N,t}^{\Delta(\alpha)}$ classifies the application as case (ii): no average effect, but evidence of lower-tail separation that requires direct QTT estimation for sign and magnitude.

tail is above or below its counterfactual benchmark. Direct QTT estimates are therefore needed to determine the sign and magnitude of the lower-tail response.

6 Conclusion

This paper proposes a pre-estimation diagnostic for distributional DID and QTT. The point is simple. Before estimating a quantile treatment effect, the researcher should first ask whether the comparison distribution is usable. A control group that is weakly or distributionally σ -diverging is not a credible benchmark for QTT. In that case, QTT should not be interpreted before the control group is sieved and the retained comparison units satisfy the mean-centered and quantile-centered non-divergence diagnostics.

Once the control group is distributionally non-diverging, the diagnostic changes its role. It no longer asks whether the benchmark can be used. It asks whether the ATT is enough. If the treated distribution is weakly and distributionally non-diverging after treatment and the ATT is zero, the diagnostic supports the no-QTT case. If the ATT is positive and constant and the treated distribution remains distributionally non-diverging, the evidence supports a common location shift. In that case, the ATT is a defensible summary of the treatment effect. If the treated-side statistics or the treated-control quantile-distance statistics reject, the ATT is not a complete description of the policy response. QTT should then be estimated at the flagged quantiles.

The interpretation of the cross-quantile statistic is deliberately limited. A rejection of $K_{N,t}^{\Delta(\alpha)}$ is evidence of separation, not a signed treatment effect. The statistic is based on squared distances. It shows that the treated α -quantile is moving away from the control benchmark, but it does not show whether the treated quantile is above or below its counterfactual quantile. For sign, magnitude, and welfare interpretation, direct QTT estimation is still required.

The asymptotic results justify the proposed decision rule. Under non-divergence, the slope statistic has a standard normal limit. Under positive weak or distributional divergence, the statistic diverges to positive infinity. The same logic applies to the cross-quantile statistic: bounded quantile gaps generate no positive slope, while growing treated-control quantile gaps produce rejection. The Monte Carlo results illustrate the same pattern in finite samples. The diagnostic keeps rejection rates low when treatment is a common shift and has high power when treatment creates growing distributional heterogeneity.

The reporting rule is therefore direct. Applied work should report the control-group non-

divergence tests, indicate whether sieving was used, report the ATT, report the treated-side and cross-quantile diagnostics, and then state the implication: mean conclusion, common-shift conclusion, or distributional conclusion requiring signed QTT. This sequence makes the maintained comparison distribution visible and prevents the ATT from being used as distributional evidence when the data show quantile separation.

7 Technical Proofs

7.1 Proof of Theorem 1

Let $G \in \{N, I\}$ denote either the control group or the treated group, and let n_G denote the number of units in group G . The data generating process is given by

$$y_{it}^N = a_i + \theta_t + e_{it}, \quad y_{it}^I = a_i + \theta_t + \beta_{it}D_{it} + e_{it},$$

The proof separates two sources of movement. The first is the usual mean-centered dispersion of group G , captured by $K_{G,t}$. The second is the distance between the group mean and the α -quantile,

$$\hat{d}_{G,t}^\alpha = \hat{\mu}_{G,t} - \hat{\mu}_{G,t}^\alpha = \mu_{G,t} - \mu_{G,t}^\alpha + (\hat{\mu}_{G,t} - \mu_{G,t}) - (\hat{\mu}_{G,t}^\alpha - \mu_{G,t}^\alpha) = \mu_{G,t} - \mu_{G,t}^\alpha + O_p(n^{-1/2}).$$

If the shape of distribution does not change over time, then the distance between the population mean and quantile is constant over time. That is, $|\mu_{G,t} - \mu_{G,t}^\alpha| = c_\alpha$. If y_{it}^G is weakly σ -converging over time, then the distance between the two decreases over time but never converge to zero. Meanwhile if y_{it}^G is diverging over time, then the distance between the two must increase over time. To combining all cases leads to

$$\hat{d}_{G,t}^\alpha = c_\alpha t^\gamma + e_{G,t}, \quad e_{G,t} = O_p(1).$$

Note that the distance between the two is bounded by $O_p(n^{-1/2})$ in the case of non-divergence, but it could be bounded by $O_p(1)$ in the case of divergence. Observe this.

$$\hat{K}_{G,t}^\alpha = \frac{1}{n_G} \sum_{i=1}^{n_G} (y_{it}^G - \hat{\mu}_{G,t}^\alpha)^2 = \frac{1}{n_G} \sum_{i=1}^{n_G} \left(y_{it}^G - \hat{\mu}_{G,t} + \hat{d}_{G,t}^\alpha \right)^2.$$

As $n_G \rightarrow \infty$, $\frac{1}{n_G} \sum_{i=1}^{n_G} (y_{it}^G - \hat{\mu}_{G,t}) = 0$, we have

$$\begin{aligned} \hat{K}_{G,t}^\alpha &= \frac{1}{n_G} \sum_{i=1}^{n_G} \alpha_{G,i}^2 + \frac{2}{n_G} \sum_{i=1}^{n_G} \alpha_{G,i} e_{G,it} + \frac{1}{n_G} \sum_{i=1}^{n_G} e_{G,it}^2 + (\hat{d}_{G,t}^\alpha)^2 \\ &= \hat{\sigma}_{\alpha,G}^2 + \hat{\sigma}_{e,G}^2 + v_{G,t} + (A_{G,t})^2 \end{aligned}$$

with $\hat{\sigma}_{\alpha,G}^2 = \frac{1}{n_G} \sum_{i=1}^{n_G} \alpha_{G,i}^2$, $\hat{\sigma}_{e,G}^2 = \frac{1}{n_G} \sum_{i=1}^{n_G} e_{G,it}^2$, $v_{G,t} = \frac{2}{n_G} \sum_{i=1}^{n_G} \alpha_{G,i} e_{G,it}$, $A_{G,t} = \hat{d}_{G,t}^\alpha = c_\alpha t^\gamma + e_{G,t}$, and $e_{G,t} = O_p(1)$. The regression used in the test is

$$\hat{K}_{G,t}^\alpha = \phi_{0,G,\alpha} + \phi_{G,\alpha} t + u_{G,t}, \quad (19)$$

Proof of Theorem 1 (i)

When $\gamma = 0$ and $e_{G,t} = O_p(1)$, $A_{G,t}^2 = c_\alpha^2 + 2c_\alpha e_{G,t} + e_{G,t}^2$. With $\gamma = 0$, the regression (19) is well specified. The constant c_α^2 is absorbed by the intercept, $\phi_{G,\alpha} = 0$ and, $u_{G,t} = v_{G,t} + a_{G,t}$, where $a_{G,t} = 2c_\alpha e_{G,t} + e_{G,t}^2$, stationary components.

The OLS of slope estimator in (19) is

$$\hat{\phi}_{G,\alpha} = \frac{\sum_{t=1}^T \tilde{t} \tilde{u}_{G,t}}{\sum_{t=1}^T \tilde{t}^2} = \frac{\sum_{t=1}^T \tilde{t} \tilde{v}_{G,t} + \sum_{t=1}^T \tilde{t} \tilde{a}_{G,t}}{\sum_{t=1}^T \tilde{t}^2}.$$

For the first term in numerator,

$$\sum_{t=1}^T \tilde{t} \tilde{v}_{G,t} = \frac{2}{n_G} \sum_{i=1}^{n_G} \tilde{\alpha}_{G,i} \sum_{t=1}^T \tilde{t} \tilde{e}_{G,it} = O_p(T^{3/2} n_G^{-1/2}) \quad \text{since} \quad \sum_{t=1}^T \tilde{t} \tilde{e}_{G,it} = O_p(T^{3/2}).$$

For the second term, define $z_{G,t} = \tilde{a}_{G,t} = a_{G,t} - \mathbb{E}[a_{G,t}]$, with $\Omega_{u,G}^2$ being the long-run variance of $z_{G,t}$. Then,

$$\text{Var} \left(\sum_{t=1}^T \tilde{t} z_{G,t} \right) = \Omega_{u,G}^2 \sum_{t=1}^T \tilde{t}^2 + o(T^3),$$

It follows that

$$\sum_{t=1}^T \tilde{t} z_{G,t} \xrightarrow{d} \mathcal{N} \left(0, \frac{T^3 \Omega_{u,G}^2}{12} \right) = O_p(T^{3/2}).$$

Therefore

$$\sum_{t=1}^T \tilde{t} \tilde{u}_{G,t} = O_p(T^{3/2} n_G^{-1/2}) + O_p(T^{3/2}) = O_p(T^{3/2}).$$

As $T, n_G \rightarrow \infty$ with $T/n_G \rightarrow 0$, the first term is negligible after $T^{-3/2}$ normalization. i.e.,

$$T^{-3/2} \sum_{t=1}^T \tilde{t} \tilde{u}_{G,t} = O_p(n_G^{-1/2}) \rightarrow 0.$$

Note that $\sum_{t=1}^T \tilde{t}z_{G,t}$ is the dominant term in the numerator which gives and, $T^{-3/2} \sum_{t=1}^T \tilde{t}z_{G,t} \xrightarrow{d} \mathcal{N}\left(0, \frac{\Omega_{u,G}^2}{12}\right)$. We get,

$$T^{3/2} \hat{\phi}_{G,\alpha} = \frac{T^{-3/2} \sum_{t=1}^T \tilde{t} \tilde{u}_{G,t}}{T^{-3} \sum_{t=1}^T \tilde{t}^2} \xrightarrow{d} \frac{\mathcal{N}\left(0, \frac{\Omega_{u,G}^2}{12}\right)}{\frac{1}{12} + o(1)} \xrightarrow{d} \mathcal{N}(0, 12\Omega_{u,G}^2)$$

Next, consider the HAC estimator

$$\hat{\Omega}_{u,G}^2 = \frac{1}{T} \sum_{t=1}^T \hat{u}_{G,t}^2 + 2 \sum_{j=1}^{\kappa} \left(1 - \frac{j}{\kappa+1}\right) \hat{\gamma}_{G,j}, \quad \hat{\gamma}_{G,j} = \frac{1}{T} \sum_{t=j+1}^T \hat{u}_{G,t} \hat{u}_{G,t-j}.$$

For estimating long run variance,

$$\frac{1}{T} \sum_{t=1}^T \hat{u}_{G,t}^2 = \frac{1}{T} \sum_{t=1}^T \left(\tilde{v}_{G,t} + \tilde{a}_{G,t} - \hat{\phi}_{G,\alpha} \tilde{t} \right)^2.$$

Expanding the square,

$$\begin{aligned} \frac{1}{T} \sum_{t=1}^T \hat{u}_{G,t}^2 &= \underbrace{\frac{1}{T} \sum_{t=1}^T \tilde{v}_{G,t}^2}_{(1)} + \underbrace{\hat{\phi}_{G,\alpha}^2 \frac{1}{T} \sum_{t=1}^T \tilde{t}^2}_{(2)} + \underbrace{\frac{1}{T} \sum_{t=1}^T \tilde{a}_{G,t}^2}_{(3)} \\ &\quad - \underbrace{2 \hat{\phi}_{G,\alpha} \frac{1}{T} \sum_{t=1}^T \tilde{t} \tilde{v}_{G,t}}_{(4)} - \underbrace{2 \hat{\phi}_{G,\alpha} \frac{1}{T} \sum_{t=1}^T \tilde{t} \tilde{a}_{G,t}}_{(5)} + \underbrace{2 \frac{1}{T} \sum_{t=1}^T \tilde{v}_{G,t} \tilde{a}_{G,t}}_{(6)}. \end{aligned}$$

The order in probability for each term can be summarized as follows. The order of the first term (1) is given by

$$\frac{1}{T} \sum_{t=1}^T \tilde{v}_{G,t}^2 = \frac{1}{T} \sum_{t=1}^T \left(\frac{2}{n_G} \sum_{i=1}^{n_G} \tilde{\alpha}_{G,i} \tilde{e}_{G,it} \right)^2 = \frac{4}{n_G^2} \sum_{i=1}^{n_G} \tilde{\alpha}_{G,i}^2 \left(\frac{1}{T} \sum_{t=1}^T \tilde{e}_{G,it}^2 \right) + o_p(n_G^{-1}) = O_p(n_G^{-1})$$

The order of the second term (2) is

$$\hat{\phi}_{G,\alpha}^2 \frac{1}{T} \sum_{t=1}^T \tilde{t}^2 = \frac{1}{T} O_p(T^{-3}) O(T^3) = O_p(T^{-1}) \quad \text{since } \hat{\phi}_{G,\alpha} = O_p(T^{-3/2}) \text{ and } \sum_{t=1}^T \tilde{t}^2 = O(T^3).$$

The order of the third term (3) goes

$$\frac{1}{T} \sum_{t=1}^T \tilde{a}_{G,t}^2 = O_p(1) \quad \text{since } a_{G,t} = 2c_\alpha e_{G,t} + e_{G,t}^2 \text{ is stationary}$$

The order of the fourth term (4) follows as

$$2\hat{\phi}_{G,\alpha} \frac{1}{T} \sum_{t=1}^T \tilde{t} \tilde{v}_{G,t} = \frac{2}{T} O_p(T^{-3/2}) O_p(T^{3/2} n_G^{-1/2}) = O_p(n_G^{-1/2} T^{-1}) \quad \text{since } \sum_{t=1}^T \tilde{t} \tilde{v}_{G,t} = O_p(T^{3/2} n_G^{-1/2}).$$

The order of the fifth term (5) becomes

$$2\hat{\phi}_{G,\alpha} \frac{1}{T} \sum_{t=1}^T \tilde{t} \tilde{a}_{G,t} = \frac{2}{T} O_p(T^{-3/2}) O_p(T^{3/2}) = O_p(T^{-1}) \quad \text{since } a_{G,t} \text{ is stationary}$$

The last term (6) is given by

$$2\frac{1}{T} \sum_{t=1}^T \tilde{v}_{G,t} \tilde{a}_{G,t} = \frac{4}{n_G} \sum_{i=1}^{n_G} \tilde{\alpha}_{G,i} \left(\frac{1}{T} \sum_{t=1}^T \tilde{e}_{G,it} \tilde{a}_{G,t} \right) = O_p(n_G^{-1/2}) O_p(T^{-1/2}) = O_p(n_G^{-1/2} T^{-1/2}).$$

As, (3) is the dominant term, we have:

$$\frac{1}{T} \sum_{t=1}^T \hat{u}_{G,t}^2 = \frac{1}{T} \sum_{t=1}^T \tilde{a}_{G,t}^2 + o_p(1).$$

For the autocovariance term,

$$\hat{u}_{G,t} = \tilde{a}_{G,t} + o_p(1),$$

so

$$\hat{\gamma}_{G,j} = \frac{1}{T} \sum_{t=j+1}^T \tilde{a}_{G,t} \tilde{a}_{G,t-j} + o_p(1) \xrightarrow{p} \gamma_{G,j}, \quad \text{where } \gamma_{G,j} = \text{Cov}(\tilde{a}_{G,t}, \tilde{a}_{G,t-j}).$$

Hence

$$\hat{\Omega}_{u,G}^2 \xrightarrow{p} \Omega_{u,G}^2, \quad \text{with } \Omega_{u,G}^2 = \mathbb{E}[\tilde{a}_{G,t}^2] + 2 \sum_{j=1}^{\kappa} \left(1 - \frac{j}{\kappa+1} \right) \gamma_{G,j}.$$

The t -statistic is

$$t_{\hat{\phi}_{G,\alpha}} = \frac{\hat{\phi}_{G,\alpha}}{\sqrt{\hat{\Omega}_{u,G}^2 / \sum_{t=1}^T \tilde{t}^2}} = \frac{T^{-3/2} \sum_{t=1}^T \tilde{t} \tilde{u}_{G,t}}{\sqrt{\hat{\Omega}_{u,G}^2 T^{-3} \sum_{t=1}^T \tilde{t}^2}}.$$

Hence, it is straightforward to show that

$$t_{\hat{\phi}_{G,\alpha}} = \frac{\hat{\phi}_{G,\alpha}}{\sqrt{12\hat{\Omega}_{u,G}^2 T^{-3}}} = \frac{T^{3/2} \hat{\phi}_{G,\alpha}}{\sqrt{12\hat{\Omega}_{u,G}^2}} \xrightarrow{d} \mathcal{N}(0, 1).$$

Proof of Theorem 1 (ii)

When $\gamma > 0$, the regression (19) is mis-specified resulting in $\phi_{G,\alpha} \neq 0$. Thus

$$\hat{\phi}_{G,\alpha} - \phi_{G,\alpha} = \frac{\sum_{t=1}^T \tilde{t} \tilde{v}_{G,t} + \sum_{t=1}^T \widetilde{\tilde{t} A_{G,t}^2}}{\sum_{t=1}^T \tilde{t}^2} \quad \text{where } A_{G,t}^2 = c_\alpha^2 t^{2\gamma} + 2c_\alpha e_{G,t} t^\gamma + e_{G,t}^2.$$

Note that

$$\sum_{t=1}^T \tilde{t} \tilde{v}_{G,t} = \frac{2}{n_G} \sum_{i=1}^{n_G} \tilde{\alpha}_{G,i} \sum_{t=1}^T \tilde{t} \tilde{e}_{G,it} = O_p(T^{3/2} n_G^{-1/2}).$$

The second component in numerator can be rewritten as

$$\sum_{t=1}^T \widetilde{\tilde{t} A_{G,t}^2} = \sum_{t=1}^T \tilde{t} A_{G,t}^2 = c_\alpha^2 \sum_{t=1}^T \tilde{t} t^{2\gamma} + 2c_\alpha \sum_{t=1}^T \tilde{t} e_{G,t} t^\gamma + \sum_{t=1}^T \tilde{t} e_{G,t}^2$$

The deterministic leading term becomes

$$\sum_{t=1}^T \tilde{t} t^{2\gamma} = T^{2\gamma+2} \int_0^1 \left(r - \frac{1}{2}\right) r^{2\gamma} dr + o(T^{2\gamma+2}).$$

Define

$$C_\gamma := \int_0^1 \left(r - \frac{1}{2}\right) r^{2\gamma} dr = \frac{1}{2\gamma+2} - \frac{1}{2(2\gamma+1)} = \frac{\gamma}{2(\gamma+1)(2\gamma+1)}.$$

Since $\gamma > 0$, $C_\gamma > 0$. Hence, $c_\alpha^2 \sum_{t=1}^T \tilde{t} t^{2\gamma} = c_\alpha^2 C_\gamma T^{2\gamma+2} + o(T^{2\gamma+2})$, $\sum_{t=1}^T \tilde{t} e_{G,t} t^\gamma = O_p(T^{\gamma+3/2})$, and $\sum_{t=1}^T \tilde{t} e_{G,t}^2 = \sum_{t=1}^T \tilde{t} z_{G,t} = O_p(T^{3/2})$.

Thus,

$$\sum_{t=1}^T \widetilde{tA_{G,t}^2} = c_\alpha^2 C_\gamma T^{2\gamma+2} + o_p(T^{2\gamma+2}).$$

As, $T, n_G \rightarrow \infty$ with $T/n_G \rightarrow 0$, the first term in the numerator goes to zero. The second term is dominant term with order $O_p(T^{2\gamma+2})$. Thus,

$$\hat{\phi}_{G,\alpha} - \phi_{G,\alpha} = \frac{\sum_{t=1}^T \tilde{v}_{G,t} + \sum_{t=1}^T \widetilde{tA_{G,t}^2}}{\sum_{t=1}^T \tilde{t}^2} = 12c_\alpha^2 C_\gamma T^{2\gamma-1} + o_p(T^{2\gamma-1}).$$

Therefore

$$\frac{\hat{\phi}_{G,\alpha}}{T^{2\gamma-1}} \xrightarrow{p} 12c_\alpha^2 C_\gamma > 0.$$

Now, we calculate the long run variance by

$$\begin{aligned} \frac{1}{T} \sum_{t=1}^T \hat{u}_t^2 &= \frac{1}{T} \sum_{t=1}^T \left(\tilde{v}_{G,t} + \widetilde{A_{G,t}^2} - \hat{\phi}_{G,\alpha} \tilde{t} \right)^2 = \underbrace{\frac{1}{T} \sum_{t=1}^T \tilde{v}_{G,t}^2}_{(1)} + \underbrace{\hat{\phi}_{G,\alpha}^2 \frac{1}{T} \sum_{t=1}^T \tilde{t}^2}_{(2)} + \underbrace{\frac{1}{T} \sum_{t=1}^T \widetilde{A_{G,t}^2}^2}_{(3)} \\ &\quad - \underbrace{2\hat{\phi}_{G,\alpha} \frac{1}{T} \sum_{t=1}^T \tilde{t} \tilde{v}_{G,t}}_{(4)} - \underbrace{2\hat{\phi}_{G,\alpha} \frac{1}{T} \sum_{t=1}^T \tilde{t} \widetilde{A_{G,t}^2}}_{(5)} + \underbrace{2\frac{1}{T} \sum_{t=1}^T \tilde{v}_{G,t} \widetilde{A_{G,t}^2}}_{(6)}. \end{aligned}$$

Each term is bounded as follow: The first term $\frac{1}{T} \sum_{t=1}^T \tilde{v}_{G,t}^2 = O_p(n_G^{-1})$. The second term becomes $\hat{\phi}_{G,\alpha}^2 \frac{1}{T} \sum_{t=1}^T \tilde{t}^2 = \frac{1}{T} O_p(T^{4\gamma-2}) \cdot O(T^3) = O_p(T^{4\gamma})$ since $\hat{\phi}_{G,\alpha} = O_p(T^{2\gamma-1})$ and $\sum_{t=1}^T \tilde{t}^2 = O(T^3)$. To find out the maximum order of the third term, consider the followings.

$$\overline{A^2} = \frac{1}{T} \sum_{t=1}^T A_{G,t}^2 = \frac{1}{T} \sum_{t=1}^T (c_\alpha^2 t^{2\gamma} + 2c_\alpha e_{G,t} t^\gamma + e_{G,t}^2).$$

Since $e_{G,t} = O_p(1)$, $\sum_{t=1}^T t^{2\gamma} = O(T^{2\gamma+1})$, and $\sum_{t=1}^T t^\gamma = O(T^{\gamma+1})$, it follows that $\overline{A^2} = O_p(T^{2\gamma}) + O_p(T^\gamma) + O_p(1) = O_p(T^{2\gamma})$. Therefore, uniformly over $t = 1, \dots, T$,

$$\widetilde{A_{G,t}^2} = A_{G,t}^2 - \overline{A^2}$$

has leading deterministic scale $T^{2\gamma}$, and the lower-order stochastic pieces from $2c_\alpha e_{G,t} t^\gamma$ and

$e_{G,t}^2$ do not change this order. Hence we have,

$$\frac{1}{T} \sum_{t=1}^T \widetilde{A_{G,t}^2}^2 = \frac{1}{T} \sum_{t=1}^T O_p(T^{4\gamma}) = O_p(T^{4\gamma}).$$

The rest of terms are bounded as

$$2\hat{\phi}_{G,\alpha} \frac{1}{T} \sum_{t=1}^T \tilde{t} \tilde{v}_{G,t} = \frac{2}{T} O_p(T^{2\gamma-1}) O_p(T^{3/2} n_G^{-1/2}) = O_p(T^{2\gamma-1/2} n_G^{-1/2}),$$

$$2\hat{\phi}_{G,\alpha} \frac{1}{T} \sum_{t=1}^T \tilde{t} \widetilde{A_{G,t}^2} = \frac{2}{T} O_p(T^{2\gamma-1}) O_p(T^{2\gamma+2}) = O_p(T^{4\gamma}),$$

and

$$2\frac{1}{T} \sum_{t=1}^T \tilde{v}_{G,t} \widetilde{A_{G,t}^2} = \frac{4}{n_G} \sum_{i=1}^{n_G} \tilde{\alpha}_{G,i} \left(\frac{1}{T} \sum_{t=1}^T \tilde{e}_{G,it} \widetilde{A_{G,t}^2} \right) = O_p(n_G^{-1/2}) O_p(T^{2\gamma}) = O_p(T^{2\gamma} n_G^{-1/2}).$$

As $T, n_G \rightarrow \infty$ and $T/n_G \rightarrow 0$, the dominant terms are (2), (3), and (5) with order $O_p(T^{4\gamma})$. Thus,

$$\frac{1}{T} \sum_{t=1}^T \hat{u}_t^2 = O_p(T^{4\gamma})$$

Consequently,

$$\hat{\Omega}_{u,G}^2 = O_p(T^{4\gamma+\kappa}).$$

$$t_{\hat{\phi}_{G,\alpha}} = \frac{\hat{\phi}_{G,\alpha} - \phi_{G,\alpha}}{\sqrt{\hat{\Omega}_{u,G}^2 / \sum_{t=1}^T \tilde{t}^2}} = \frac{O_p(T^{2\gamma-1})}{\sqrt{O_p(T^{4\gamma+\kappa})/O(T^3)}} = \frac{O_p(T^{2\gamma-1})}{O(T^{2\gamma-3/2+\kappa/2})} = O_p(T^{1/2-\kappa/2}).$$

Because $0 < \kappa < 1$, the exponent $1/2 - \kappa/2$ is positive. In particular, for $\kappa = 1/3$,

$$1/2 - \kappa/2 = 1/3, \quad t_{\hat{\phi}_{G,\alpha}} = O_p(T^{1/3}).$$

As $T, n_G \rightarrow \infty$ and $T/n_G \rightarrow 0$ and $\phi_{G,\alpha} > 0$,

$$t_{\hat{\phi}_{G,\alpha}} \rightarrow +\infty.$$

Proof of Theorem 2

Let

$$\hat{\Delta}_t^\alpha = \hat{\mu}_{I,t}^\alpha - \hat{\mu}_{N,t}^\alpha = c_\alpha^\Delta t^\gamma + \epsilon_t,$$

where $\epsilon_t = O_p(n^{-1/2})$ if $\gamma \leq 0$, but $\epsilon_t = O_p(1)$ if $\gamma > 0$.

The non-divergent case is $\gamma = 0$, corresponding to a stable treated-control quantile gap. The divergent cross-group case is $\gamma > 0$, corresponding to a growing quantile gap generated by time-varying treatment effects.

Consider the cross-group statistic

$$\hat{K}_{N,t}^{\Delta(\alpha)} = \frac{1}{n_N} \sum_{i=1}^{n_N} (y_{it}^N - \hat{\mu}_{I,t}^\alpha)^2.$$

Substituting $\hat{\mu}_{I,t}^\alpha = \hat{\mu}_{N,t}^\alpha + \hat{\Delta}_t^\alpha$ and simplifying gives,

$$\begin{aligned} \hat{K}_{N,t}^{\Delta(\alpha)} &= \frac{1}{n_N} \sum_{i=1}^{n_N} (y_{it}^N - \hat{\mu}_{N,t}^\alpha)^2 + \left(\hat{\Delta}_t^\alpha\right)^2 - 2\hat{\Delta}_t^\alpha \frac{1}{n_N} \sum_{i=1}^{n_N} (y_{it}^N - \hat{\mu}_{N,t}^\alpha) \\ &= \hat{K}_{N,t}^\alpha + \left(\hat{\Delta}_t^\alpha\right)^2 - 2\hat{\Delta}_t^\alpha e_{N,t} \\ &= \hat{\sigma}_{\alpha,N}^2 + \hat{\sigma}_{e,N}^2 + v_{N,t} + e_{N,t}^2 + (\hat{\Delta}_t^\alpha)^2 - 2\hat{\Delta}_t^\alpha e_{N,t}, \\ &= \hat{\sigma}_{\alpha,N}^2 + \hat{\sigma}_{e,N}^2 + v_{N,t} + r_t. \end{aligned}$$

$$\text{with } \hat{\sigma}_{\alpha,N}^2 = \frac{1}{n_N} \sum_{i=1}^{n_N} \alpha_{N,i}^2, \hat{\sigma}_{e,N}^2 = \frac{1}{n_N} \sum_{i=1}^{n_N} e_{N,ii}^2, v_{N,t} = \frac{2}{n_N} \sum_{i=1}^{n_N} \alpha_{N,i} e_{N,ii}, r_t = e_{N,t}^2 + \left(\hat{\Delta}_t^\alpha\right)^2 - 2\hat{\Delta}_t^\alpha e_{N,t}, e_{N,t} = \hat{d}_{N,t}^\alpha = \hat{\mu}_{N,t} - \hat{\mu}_{N,t}^\alpha = O_p(1)$$

The trend regression is

$$\hat{K}_{N,t}^{\Delta(\alpha)} = \phi_{0,\Delta} + \phi_{N,\Delta,\alpha} t + u_{\Delta,t}, \tag{20}$$

where $u_{\Delta,t} = v_{N,t} + r_t$. Throughout this proof, we assume that as $T, n_N \rightarrow \infty$, $T/n_N \rightarrow 0$.

Proof of Theorem 2 (i)

When $\gamma = 0$, the trend regression in (20) is well specified. The true value of $\phi_{N,\Delta,\alpha}$ equals zero. Hence it is straightforward to show that as $T, n_N \rightarrow \infty$ with $T/n_N \rightarrow 0$,

$$T^{3/2} \hat{\phi}_{N,\Delta,\alpha} \xrightarrow{d} \mathcal{N}(0, 12\Omega_\Delta^2).$$

Also, its t -ratio has the following limiting distribution.

$$t_{\hat{\phi}_{N,\Delta,\alpha}} = \frac{\hat{\phi}_{N,\Delta,\alpha}}{\sqrt{\hat{\Omega}_\Delta^2 / \sum_{t=1}^T \tilde{t}^2}} \xrightarrow{d} \mathcal{N}(0, 1).$$

Proof of Theorem 2(ii)

As long as c_α and c_α^Δ are constant, the proof of Theorem 2 (ii) is exactly same as that of Theorem 1 (ii). Hence it is omitted here.

7.2 Proof of Corollary 1

Let

$$\hat{d}_{G,t}^\alpha = \hat{\mu}_{G,t} - \hat{\mu}_{G,t}^\alpha, \quad G \in \{I, N\}.$$

Then

$$\hat{\mu}_{G,t}^\alpha = \hat{\mu}_{G,t} - \hat{d}_{G,t}^\alpha.$$

Therefore,

$$\widehat{QTT}_t^\alpha = \hat{\mu}_{I,t}^\alpha - \hat{\mu}_{N,t}^\alpha = (\hat{\mu}_{I,t} - \hat{d}_{I,t}^\alpha) - (\hat{\mu}_{N,t} - \hat{d}_{N,t}^\alpha). \quad (\text{C.1})$$

Let

$$\widehat{ATT}_t = \hat{\mu}_{I,t} - \hat{\mu}_{N,t} = \hat{\Delta}_t.$$

Then equation (C.1) gives

$$\widehat{QTT}_t^\alpha = \hat{\Delta}_t - (\hat{d}_{I,t}^\alpha - \hat{d}_{N,t}^\alpha). \quad (\text{C.2})$$

By the pre-treatment quantile-equality restriction and regularity of the sample mean and quantile estimators,

$$\hat{d}_{I,t}^\alpha - \hat{d}_{N,t}^\alpha = O_p(n_{\min}^{-1/2}), \quad t \in \mathcal{T}_0, \quad \alpha \in \mathcal{A}, \quad (\text{C.3})$$

where

$$n_{\min} = \min\{n_I, n_N\}.$$

By Assumption A2, the retained control group is distributionally non-diverging. By the maintained condition of Corollary 1, the post-treatment treated group is also weakly and distributionally non-diverging:

$$t_{\hat{\phi}_I} \leq 1.65, \quad \sup_{\alpha \in \mathcal{A}} t_{\hat{\phi}_{I,\alpha}} \leq 1.65. \quad (\text{C.4})$$

Thus, in case (i), the diagnostics do not reject post-treatment quantile-specific divergence in the treated group. Together with the pre-treatment equality in (C.3), this implies that the estimated treated mean-to-quantile distance does not separate from the retained control mean-to-quantile distance after treatment:

$$\hat{d}_{I,t}^\alpha - \hat{d}_{N,t}^\alpha = O_p(n_{\min}^{-1/2}), \quad t \in \mathcal{T}_1, \quad \alpha \in \mathcal{A}. \quad (\text{C.5})$$

Substituting (C.5) into (C.2) gives

$$\widehat{QTT}_t^\alpha = \hat{\Delta}_t + O_p(n_{\min}^{-1/2}) = \widehat{ATT}_t + O_p(n_{\min}^{-1/2}). \quad (\text{C.6})$$

If $ATT_t = \Delta_t = 0$, then

$$\widehat{ATT}_t = O_p(n_{\min}^{-1/2}). \quad (\text{C.7})$$

Using (C.6),

$$\widehat{QTT}_t^\alpha = O_p(n_{\min}^{-1/2}), \quad \alpha \in \mathcal{A}. \quad (\text{C.8})$$

If $ATT_t = \Delta_t \neq 0$ and Assumption A4 holds, then

$$\Delta_t - \Delta_{t-1} = 0, \quad t, t-1 \in \mathcal{T}_1. \quad (\text{C.9})$$

Therefore the population average treatment effect is constant over the post-treatment period. By regularity of the sample mean estimators,

$$\widehat{ATT}_t = \Delta_t + O_p(n_{\min}^{-1/2}). \quad (\text{C.10})$$

Using (C.6) and (C.10),

$$\widehat{QTT}_t^\alpha = \Delta_t + O_p(n_{\min}^{-1/2}), \quad \alpha \in \mathcal{A}. \quad (\text{C.11})$$

Equations (C.8) and (C.11) imply the two sample conclusions in Corollary 1: under zero ATT, the estimated QTT is asymptotically zero, and under constant nonzero ATT, the estimated QTT equals the ATT up to sampling error.

Equivalently,

$$\widehat{QTT}_t^\alpha - \widehat{ATT}_t = O_p(n_{\min}^{-1/2}), \quad \alpha \in \mathcal{A}. \quad (\text{C.12})$$

Thus, in case (i), the ATT implies QTT.

7.3 Proof of Corollary 2

For some $\alpha \in \mathcal{A}$, suppose the sample cross-quantile separation alternative holds:

$$\widehat{\Delta}_t^\alpha = \widehat{\Delta}_t + c_\alpha^\Delta t^\gamma + \epsilon_t + O_p(n_{\min}^{-1/2}), \quad c_\alpha^\Delta \neq 0, \quad \gamma > 0, \quad \epsilon_t = O_p(1), \quad (\text{C.13})$$

where

$$\widehat{\Delta}_t = \widehat{ATT}_t, \quad n_{\min} = \min\{n_I, n_N\}.$$

By Assumption A2, the retained control group is distributionally non-diverging, so

$$\widehat{K}_{N,t}^\alpha = O_p(1), \quad \widehat{e}_{N,t} = O_p(1). \quad (\text{C.14})$$

The sample cross-quantile statistic satisfies the decomposition

$$\widehat{K}_{N,t}^{\Delta(\alpha)} = \widehat{K}_{N,t}^\alpha + \left(\widehat{\Delta}_t^\alpha\right)^2 - 2\widehat{\Delta}_t^\alpha \widehat{e}_t. \quad (\text{C.15})$$

First consider case (ii), where

$$ATT_t = \Delta_t = 0.$$

Then

$$\widehat{\Delta}_t = O_p(n_{\min}^{-1/2}),$$

and (C.13) becomes

$$\widehat{\Delta}_t^\alpha = c_\alpha^\Delta t^\gamma + \epsilon_t + O_p(n_{\min}^{-1/2}). \quad (\text{C.16})$$

Substituting (C.16) into (C.15),

$$\begin{aligned} \widehat{K}_{N,t}^{\Delta(\alpha)} &= \widehat{K}_{N,t}^\alpha + (c_\alpha^\Delta)^2 t^{2\gamma} + 2c_\alpha^\Delta t^\gamma \epsilon_t + \epsilon_t^2 \\ &\quad - 2c_\alpha^\Delta t^\gamma \widehat{\epsilon}_t - 2\epsilon_t \widehat{\epsilon}_t + O_p(t^\gamma n_{\min}^{-1/2}) + O_p(n_{\min}^{-1}). \end{aligned}$$

Using (C.14),

$$\widehat{K}_{N,t}^{\Delta(\alpha)} = (c_\alpha^\Delta)^2 t^{2\gamma} + O_p(t^\gamma) + O_p(1) + O_p(t^\gamma n_{\min}^{-1/2}). \quad (\text{C.17})$$

Next consider case (iii), where Assumption A4 gives

$$ATT_t = \Delta_t = \Delta \neq 0, \quad \Delta_t - \Delta_{t-1} = 0.$$

Then

$$\widehat{\Delta}_t = \Delta + O_p(n_{\min}^{-1/2}),$$

and

$$\widehat{\Delta}_t^\alpha = \Delta + c_\alpha^\Delta t^\gamma + \epsilon_t + O_p(n_{\min}^{-1/2}). \quad (\text{C.18})$$

Therefore,

$$\begin{aligned} (\widehat{\Delta}_t^\alpha)^2 &= (c_\alpha^\Delta)^2 t^{2\gamma} + 2c_\alpha^\Delta t^\gamma (\Delta + \epsilon_t) + (\Delta + \epsilon_t)^2 \\ &\quad + O_p(t^\gamma n_{\min}^{-1/2}) + O_p(n_{\min}^{-1/2}). \end{aligned}$$

Substitution into (C.15), together with (C.14), yields

$$\widehat{K}_{N,t}^{\Delta(\alpha)} = (c_\alpha^\Delta)^2 t^{2\gamma} + O_p(t^\gamma) + O_p(1) + O_p(t^\gamma n_{\min}^{-1/2}). \quad (\text{C.19})$$

Thus, in both cases (ii) and (iii),

$$\widehat{K}_{N,t}^{\Delta(\alpha)} = (c_\alpha^\Delta)^2 t^{2\gamma} + O_p(t^\gamma) + O_p(1) + O_p(t^\gamma n_{\min}^{-1/2}). \quad (\text{C.20})$$

Under the rate condition $T/n_{\min} \rightarrow 0$, the sampling-error term is asymptotically negligible relative to the leading deterministic component. Therefore,

$$\widehat{K}_{N,t}^{\Delta(\alpha)} = (c_\alpha^\Delta)^2 t^{2\gamma} + O_p(t^\gamma) + O_p(1). \quad (\text{C.21})$$

Let

$$C_\gamma = \int_0^1 \left(r - \frac{1}{2}\right) r^{2\gamma} dr = \frac{\gamma}{2(\gamma+1)(2\gamma+1)} > 0.$$

Then

$$\sum_{t=1}^T \tilde{t} \widehat{K}_{N,t}^{\Delta(\alpha)} = (c_\alpha^\Delta)^2 C_\gamma T^{2\gamma+2} + o_p(T^{2\gamma+2}),$$

and

$$\widehat{\phi}_{N,\Delta,\alpha} = 12(c_\alpha^\Delta)^2 C_\gamma T^{2\gamma-1} + o_p(T^{2\gamma-1}). \quad (\text{C.22})$$

By Appendix A.2, as $T, n_{\min} \rightarrow \infty$ with $T/n_{\min} \rightarrow 0$,

$$t_{\widehat{\phi}_{N,\Delta,\alpha}} \rightarrow +\infty, \quad \Pr\left(t_{\widehat{\phi}_{N,\Delta,\alpha}} > 1.65\right) \rightarrow 1. \quad (\text{C.23})$$

If $ATT_t = 0$, then

$$\widehat{QTT}_t^\alpha = \widehat{\Delta}_t^\alpha = c_\alpha^\Delta t^\gamma + \epsilon_t + O_p(n_{\min}^{-1/2}),$$

so

$$\widehat{QTT}_t^\alpha \neq O_p(n_{\min}^{-1/2}). \quad (\text{C.24})$$

Thus the estimated quantile effect is not summarized by the zero ATT.

If $ATT_t = \Delta \neq 0$, then

$$\widehat{QTT}_t^\alpha - \widehat{ATT}_t = \widehat{\Delta}_t^\alpha - \widehat{\Delta}_t = c_\alpha^\Delta t^\gamma + \epsilon_t + O_p(n_{\min}^{-1/2}),$$

so

$$\widehat{QTT}_t^\alpha - \widehat{ATT}_t \neq O_p(n_{\min}^{-1/2}). \quad (\text{C.25})$$

Thus the estimated quantile effect is not summarized by the constant nonzero ATT.

Equations (C.23)–(C.25) imply that, for every flagged quantile, QTT should be estimated directly whether the ATT is zero or constant nonzero. Since $\widehat{K}_{N,t}^{\Delta(\alpha)}$ is a squared-distance statistic, the rejection indicates quantile separation, not the sign or magnitude of the quantile effect.

7.4 Sieving Algorithm

This section describes the sieving algorithm used in the paper. The goal is to construct a retained set of units whose outcome paths are non-diverging both around their mean and

around the relevant quantiles. The pre-treatment sieve is applied to the combined treated-control candidate set over $\mathcal{T}_0$. The all-period control sieve is then applied to the retained control units over $t = 1, \dots, T$.

Let $\mathcal{S}_0$ denote the initial candidate set and let $\mathcal{T}_S$ denote the time window for the sieve. For any candidate set $\mathcal{S}$ with N_S units, define

$$\hat{\mu}_{S,t} = \frac{1}{N_S} \sum_{i \in \mathcal{S}} y_{it}, \quad \hat{\mu}_{S,t}^\alpha = \text{the } \alpha\text{-quantile of } \{y_{it} : i \in \mathcal{S}\}.$$

The set-level mean-centered and quantile-centered dispersion statistics are

$$\hat{K}_{S,t} = \frac{1}{N_S} \sum_{i \in \mathcal{S}} (y_{it} - \hat{\mu}_{S,t})^2,$$

where n_S is the number of units in a set $\mathcal{S}$ and

$$\hat{K}_{S,t}^\alpha = \frac{1}{N_S} \sum_{i \in \mathcal{S}} (y_{it} - \hat{\mu}_{S,t}^\alpha)^2, \quad \alpha \in \mathcal{A}.$$

For each candidate set, run

$$\hat{K}_{S,t} = \phi_{S,0} + \phi_S t + u_{S,t}, \quad t \in \mathcal{T}_S,$$

and

$$\hat{K}_{S,t}^\alpha = \phi_{S,0,\alpha} + \phi_{S,\alpha} t + u_{S,t}^\alpha, \quad t \in \mathcal{T}_S, \quad \alpha \in \mathcal{A}.$$

The candidate set passes the sieve if

$$t_{\hat{\phi}_S} \leq 1.65, \quad \sup_{\alpha \in \mathcal{A}} t_{\hat{\phi}_{S,\alpha}} \leq 1.65. \quad (\text{S.1})$$

Thus a retained set must pass both the mean-centered and quantile-centered non-divergence tests.

Pre-treatment sieve

For the pre-treatment period, set $\mathcal{T}_S = \mathcal{T}_G$ and let $\mathcal{S}_G$ be the set of either ever-treated group(I) or control group(N) observed before treatment. Let $\mathcal{N}_{SG}$ be the number of units in

group (I, N) before treatment. The pre-treatment sieve has two components.

1. Mean-centered sieving. For each $i \in \mathcal{S}_0$, define

$$s_{it}^\mu = y_{it} - \hat{\mu}_{\mathcal{S}_G, t}, \quad t \in \mathcal{T}_G,$$

and run the unit-level regression

$$(s_{it}^\mu)^2 = \varphi_{i,0}^\mu + \varphi_i^\mu t + \eta_{it}^\mu, \quad t \in \mathcal{T}_0.$$

Let $t_{\hat{\varphi}_i^\mu}$ be the HAC t -statistic for $\hat{\varphi}_i^\mu$. The mean-centered core group is

$$\mathcal{G}_{\mu,0} = \left\{ i \in \mathcal{S}_G : t_{\hat{\varphi}_i^\mu} < -1.2 \right\}. \quad (\text{S.2})$$

This step removes units whose squared distance from the pre-treatment cross-sectional mean has a positive trend.

2. Quantile-centered sieving. For each $i \in \mathcal{S}_G$ and $\alpha \in \mathcal{A}$, define

$$s_{it}^\alpha = y_{it} - \hat{\mu}_{\mathcal{S}_G, t}^\alpha, \quad t \in \mathcal{T}_0,$$

and run

$$(s_{it}^\alpha)^2 = \varphi_{i,0}^\alpha + \varphi_i^\alpha t + \eta_{it}^\alpha, \quad t \in \mathcal{T}_0, \quad \alpha \in \mathcal{A}.$$

Let $t_{\hat{\varphi}_i^\alpha}$ be the HAC t -statistic for $\hat{\varphi}_i^\alpha$. The quantile-centered core group is

$$\mathcal{G}_{\alpha,0} = \left\{ i \in \mathcal{S}_G : \sup_{\alpha \in \mathcal{A}} t_{\hat{\varphi}_i^\alpha} < -1.2 \right\}. \quad (\text{S.3})$$

This step removes units whose squared distance from at least one pre-treatment quantile has a positive trend.

The initial pre-treatment core group is the set that passes both screens:

$$\mathcal{G}_0 = \mathcal{G}_{\mu,0} \cap \mathcal{G}_{\alpha,0}. \quad (\text{S.4})$$

Let $N_{\mathcal{G}_0}$ be the number of units in the pre-treatment core group $\mathcal{G}_0$. The remaining units are

$$\mathcal{G}_0^c = \mathcal{S}_G \setminus \mathcal{G}_0.$$

3. Depth ranking and enrichment. For each $j \in \mathcal{G}_0^c$, construct

$$M_j = (\bar{m}_j, \bar{q}_j(\alpha_1), \dots, \bar{q}_j(\alpha_m))',$$

where $\{\alpha_1, \dots, \alpha_m\} \subset \mathcal{A}$ is the quantile grid and

$$\bar{m}_j = \frac{1}{T_0} \sum_{t \in \mathcal{T}_0} (y_{jt} - \hat{\mu}_{\mathcal{G}_0, t})^2, \quad \bar{q}_j(\alpha) = \frac{1}{T_0} \sum_{t \in \mathcal{T}_0} (y_{jt} - \hat{\mu}_{\mathcal{G}_0, t}^\alpha)^2.$$

Let

$$\hat{\mu}_{G, M} = \frac{1}{N_{G0}} \sum_{i \in \mathcal{G}_0} M_i, \quad \hat{\Sigma}_M = \frac{1}{N_{G0} - 1} \sum_{i \in \mathcal{G}_0} (M_i - \hat{\mu}_{G, M})(M_i - \hat{\mu}_{G, M})'.$$

For $j \in \mathcal{G}_0^c$, define

$$O_j = (M_j - \hat{\mu}_{G, M})' \hat{\Sigma}_M^{-1} (M_j - \hat{\mu}_{G, M}), \quad D_j = \frac{1}{1 + O_j}.$$

Order units in $\mathcal{G}_0^c$ from largest to smallest D_j . Starting from $\mathcal{G}_0$, add units one at a time according to this ranking. After each addition, recompute $\hat{K}_{\mathcal{S}, t}$ and $\hat{K}_{\mathcal{S}, t}^\alpha$ over $t \in \mathcal{T}_0$ and keep the enlarged set only if (S.1) holds. The resulting retained pre-treatment set is denoted by $\hat{\mathcal{G}}_0$.

All-period control sieve

After the pre-treatment sieve, restrict attention to the retained control units and set $\mathcal{T}_S = \{1, \dots, T\}$. Repeat the same mean-centered and quantile-centered steps using only control units. The final retained control group is denoted by $\hat{\mathcal{G}}_N$. This is the group used to compute

$$\hat{K}_{N, t}, \quad \hat{K}_{N, t}^\alpha, \quad \hat{K}_{N, t}^{\Delta(\alpha)}.$$

The control benchmark is admissible for QTT only if

$$t_{\hat{\phi}_N} \leq 1.65, \quad \sup_{\alpha \in \mathcal{A}} t_{\hat{\phi}_{N, \alpha}} \leq 1.65.$$

References

- Athey, S. and G. W. Imbens (2006). Identification and inference in nonlinear difference-in-differences models. *Econometrica* 74(2), 431–497.
- Bosch, M. and R. M. Campos-Vazquez (2014). The trade-offs of welfare policies in labor markets with informal jobs: The case of the Seguro Popular program in Mexico. *American Economic Journal: Economic Policy* 6(4), 71–99.
- Borusyak, K., X. Jaravel, and J. Spiess (2024). Revisiting event-study designs: Robust and efficient estimation. *Review of Economic Studies* 91(6), 3253–3285.
- Callaway, B. and T. Li (2019). Quantile treatment effects in difference-in-differences models with panel data. *Quantitative Economics* 10(4), 1579–1618.
- Callaway, B., T. Li, and T. Oka (2018). Quantile treatment effects in difference-in-differences models under dependence restrictions and with only two time periods. *Journal of Econometrics* 206(2), 395–413.
- Callaway, B. and P. H. C. Sant’Anna (2021). Difference-in-differences with multiple time periods. *Journal of Econometrics* 225(2), 200–230.
- Chernozhukov, V., I. Fernández-Val, and B. Melly (2013). Inference on counterfactual distributions. *Econometrica* 81(6), 2205–2268.
- Galvao, A. F. and B. I. Miller (2025). Doubly-robust quantile treatment effects with staggered interventions. Manuscript.
- Kim, D. and J. M. Wooldridge (2025). Difference-in-differences estimator of quantile treatment effect on the treated. *Journal of Business & Economic Statistics* 43(2), 401–412.
- Kong, J., P. C. B. Phillips, and D. Sul (2019). Weak σ -convergence: Theory and applications. *Journal of Econometrics* 209(2), 185–207.
- Lee, Y. and D. Sul (2026). Powerful parallel trends test and sieving algorithm for identifying control units satisfying parallel trends. *mimeo* University of Texas at Dallas (2026).
- Lee, Y., P. C. B. Phillips, S. Song, and D. Sul (2026). Identifying Common Trend Determinants in Panel Data. *mimeo* University of Texas at Dallas.
- Li, X. and L. Lin (2024). Identification of quantile treatment effects in difference-in-

- differences settings with staggered adoption. *Economics Letters* 241, 111792.
- Phillips, P. C. B. and D. Sul (2007). Transition modeling and econometric convergence tests. *Econometrica* 75(6), 1771–1855.
- Roth, J. and P. H. C. Sant’Anna (2023). When is parallel trends sensitive to functional form? *Econometrica* 91(2), 737-747.
- Roth, J., P. H. C. Sant’Anna, A. Bilinski, and J. Poe (2023). What’s trending in difference-in-differences? A synthesis of the recent econometrics literature. *Journal of Econometrics* 235(2), 2218–2244.
- Sant’Anna, P. H. C. and J. Zhao (2020). Doubly robust difference-in-differences estimators. *Journal of Econometrics* 219(1), 101–122.